

Room correction by inversion in REW — a step-by-step guide

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Room correction by inversion in REW

A step-by-step guide, with the reasoning behind each choice.

About this document

This is a **procedure**, not a lab notebook. Every number in it has been measured, but the measurements, the false starts and the corrections live in `NOTES.md`; nothing of that is repeated here. If a statement here disagrees with `NOTES.md`, this document is the later one and wins.

Scope. Producing a pair of FIR room-correction filters for a stereo pair, in REW, by **inverting the measured response** — as opposed to fitting parametric filters with Auto EQ. The filters are minimum phase, exported as WAV, and convolved by BruteFIR.

The configuration assumed throughout is DRC-120.blue, which is the current one:

Loudspeakers	B&W Nautilus 801 , 3-way, crossovers at 350 Hz and 4 kHz, third order
Placement	120 cm from the front wall; microphone 449 cm from the front wall
Room	$\approx 58 \text{ m}^3$, irregular (7.40 m long, 4.186 $\rightarrow$ 5.986 m wide, ceiling 2.4 $\rightarrow$ 3.0 m, open corridor)
Schroeder frequency	$\approx$ 166 Hz
Correction band	20 – 225 Hz
Sample rate	48 kHz throughout
Crossover correction	X801.wav, built in rePhase, 131072 taps
Convolution engine	BruteFIR
Data	../DRC-120.blue/120-blue-with-inversion.mdat and the exports in ../DRC-120.blue/120.blue-with-inversion.txts/

Not covered: subwoofers, speaker placement, Auto EQ, room treatment.

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Part I — What inversion actually does

1. The method in one paragraph

Measure the loudspeaker in the room. Decide what you *want* that response to be — the **target**. Divide the target by the measurement. The quotient is a correction curve: wherever the room was 6 dB loud the quotient is −6 dB, wherever it was 4 dB shy the quotient is +4 dB. Turn that curve into an impulse response and convolve it with the music. Because convolution in time is multiplication in frequency, **measurement** × **correction** = **target** — the room is cancelled by construction.

That is the whole idea, and it is exact. Everything difficult about the method comes from one place: **the division is an ill-posed inverse**. Where the measurement is very small, the quotient is very large; where the measurement has a razor-thin notch, the quotient has a razor-thin resonance. An unguarded inversion faithfully reproduces every defect of the measurement, including the ones that are not properties of the room at all.

So the whole craft of this method is deciding what the inversion is allowed to see. Every setting in this guide is a form of that decision.

2. What it can correct, and what it provably cannot

Any measured response factors — uniquely — into two parts:

$$H(\text{measured}) = H(\text{minimum phase}) \times H(\text{excess phase})$$

- **H(min)** carries all the magnitude. Its phase is not free: it is fixed by the magnitude through the Hilbert relation. Know the magnitude and you know the phase.
- **H(excess)** has magnitude exactly 1 everywhere. It is pure phase — pure delay and rotation, invisible in an amplitude plot.

This procedure produces a **minimum-phase** filter (that is what steps 4 and 8 enforce). A minimum-phase filter is derived from a magnitude. Therefore:

A minimum-phase inversion corrects H(min) completely and H(excess) not at all — not “poorly”, but exactly zero.

That is not a limitation to be engineered around; it is the point. Look at what **H(excess)** contains here:

component of the excess phase	invert it?	why
bulk propagation delay	no	it is the speed of sound; inaudible, and inverting it is acausal
room reflections	no	position-specific, changes over 10 cm, and its inverse is acausal — it pre-rings
crossover all-pass	yes	a property of the loudspeaker; position-independent; deterministic; known in closed form

Two of the three must be left alone. The one that should be corrected is the one the minimum-phase route cannot reach. **That is the entire job of X801**, and it is why the crossover correction is a separate, model-based filter rather than something the inversion discovers.

Restricting the inversion to minimum phase is therefore a choice, not a shortcoming. A **complex** inversion — dividing by the raw measured response, phase and all — would correct all three rows of that table, including the two that must not be corrected. It would invert the room’s reflections, which needs an acausal filter, pre-rings, and is valid only at the microphone point. Steps 4 and 8 exist to make sure that never happens. The price of that safety is the third row, and X801 is what pays it.

3. Do you still need X801? Yes — and here is the proof

The question is fair: if inversion corrects the response, shouldn’t it correct the phase everywhere too, by definition?

It would — if it were a **complex** inversion. It is not: steps 4 and 8 force the filter to minimum phase, and §2 gives the reason — a complex inversion would also invert the room’s reflections, which must not be inverted. And the crossover’s contribution is precisely the kind of thing a minimum-phase inversion is blind to:

A third-order Butterworth crossover sums to

$$\frac{1}{(s+1)(s^2+s+1)} + \frac{s^3}{(s+1)(s^2+s+1)} = \frac{s^2 - s + 1}{s^2 + s + 1}$$

— a **second-order all-pass**. Magnitude 1.000000 across the whole band, 359° of phase rotation. A 4th-order Linkwitz-Riley sums to the same form with Q = 0.707 instead of 1.0. **Any crossover that sums to flat magnitude puts its phase where magnitude-derived correction cannot see it.**

Measured on the actual file, over the whole band 5 Hz – 23 kHz:

|X801.wav| min -0.00000 dB max +0.00000 dB

And the decisive test — take the minimum-phase version of X801.wav:

	result
peak	1.000000 at sample 0
energy anywhere but sample 0	2.96×10^{-14} %

The minimum-phase copy of X801 is a unit impulse. Not approximately — to fourteen decimal places. Ask the inversion to correct the crossover and it correctly answers “there is nothing here”.

The answer, stated plainly

Yes, you still need X801, and no amount of inversion will ever replace it. They address disjoint parts of the response — magnitude and magnitude-implied phase on one side, crossover all-pass on the other — and because X801 has exactly flat magnitude, cascading them cannot double-correct anything. See [R7](#).

(Could a full complex inversion do both at once? Yes, and it would also try to invert the room’s excess phase — position-specific, acausal, pre-ringing. Separating loudspeaker excess phase from room excess phase needs frequency-dependent windowing of the kind DRC-FIR, Acourate and Audiolense do. That is a different tool, not a REW setting.)

4. The one law you cannot escape

$$\Delta f \cdot \Delta t \approx 1$$

A feature Δf wide in frequency takes about $1/\Delta f$ seconds to unfold in time. It is not a rule of thumb, it is the Fourier transform. It appears in this procedure in four disguises, and they are all the same statement:

you choose	you have thereby chosen
frequency resolution of the correction curve	the length of the filter’s ringing
the length of the IR window	the frequency resolution
the number of FDW cycles	the highest Q the correction may contain
smoothing bandwidth	the highest Q the correction may contain

Made concrete at 50 Hz:

resolution of the correction curve	width at 50 Hz	shortest possible ringing
1 FFT bin (no smoothing at all)	0.37 Hz	2731 ms
1/48 octave — REW’s “Var” below 100 Hz	0.72 Hz	1385 ms
1/12 octave	2.89 Hz	346 ms
1/6 octave	5.78 Hz	173 ms
1/3 octave	11.58 Hz	86 ms

Read the right-hand column as “*how long the woofer keeps moving after the music stops*”. **You do not get to choose the correction’s resolution and its ringing separately. They are one number.**

This is why “just correct everything accurately” is not available. The finer the correction, the longer the filter rings — and beyond some point you are trading an inaudible frequency-response error for an audible time-domain one.

5. The failure this guide exists to prevent

Run the inversion with defaults and this is what you get. It happened here, in September 2025, and the filters were in service for months.

The left channel’s measurement contains a **38.3 dB null at 28.93 Hz** — a razor-thin canyon, 2 FFT bins wide. It is not a room mode: the right channel varies only 6.4 dB over the same span, so it is one loudspeaker’s path to one microphone point. Nothing was regularised, so the inversion did its job faithfully and built the exact inverse: a **Q = 40** resonator.

what the deployed filter did

narrowest feature below 200 Hz	0.8 FFT bins
group delay excursion, 20–200 Hz	80 ms (a correction filter should show a few ms)
28.7 Hz gated note, time to fall 40 dB	1348 ms

Audible as *the music stops and the woofers keep moving*. And the tell that it was never real: a Q of 40 at 29 Hz implies a 3.07 second decay. In 58 m³ that is physically impossible. **The filter was correcting something that does not exist.**

Every non-default setting in Part III is aimed at this one failure.

Part II — Decisions to take before you measure

6. One microphone position, or several?

Multiple positions is the only change on the list that attacks the cause rather than the symptom. Everything else in this guide suppresses narrow features on the grounds that they are *probably* not real. Spatial averaging finds out.

Why it works, and why it works especially well here

A genuine room mode is a property of the volume: a 50 Hz mode is at 50 Hz everywhere in the room, and only its *amplitude* varies with position. An interference null is a property of a path: two arrivals cancelling at one point, and it moves or vanishes when you move.

Average several positions and modes survive while interference washes out. That is a **discrimination no amount of smoothing can perform** — smoothing blurs everything equally, because a single measurement contains no information about which features are which.

And deep nulls are the most position-sensitive feature there is. A 38 dB null requires two arrivals to cancel to within a fraction of a decibel; move 20 cm and that balance is gone. So even a modest spread attacks exactly the feature that causes the damage.

The caveats, honestly

1. **Use the RMS average. Never the vector average.** REW’s help: > “**Vector average** ... averages the currently selected traces taking into > account both magnitude and phase. It ... is most appropriate for multiple > measurements taken from the same position, or measurements which have been > time and level aligned.”

Vector-averaging *different* positions lets the position-dependent phase cancel and **manufactures new nulls** — the exact disease you are trying to cure. The **RMS average** is a power average, so a null at one position is filled by the others; that is the behaviour you want.

2. **Level-align first.** REW again: > “If the measurements were made at different positions (*spatial averaging*) > it is usually best to first use the **Align SPL...** feature to remove > overall level differences due to different source distances.”
3. **Spacing buys you less at low frequency than you would like.** Two points decorrelate when they are a useful fraction of a wavelength apart:

	30 Hz	50 Hz	100 Hz	200 Hz
λ	11.4 m	6.9 m	3.4 m	1.7 m
$\lambda/4$	2.9 m	1.7 m	0.86 m	0.43 m

A ± 30 cm cluster around one seat decorrelates well above ~ 250 Hz, partly from 100–250 Hz, and hardly at all below 60 Hz. **Spatial averaging does not solve the bass by itself.** It still helps below 60 Hz — via the null-depth sensitivity above — but do not expect the 30 Hz region to be cleaned up by moving the mic 30 cm.

4. **It is not free.** 5 positions $\times$ 2 channels = 10 sweeps, and one contaminated position (mic bumped, door opened) poisons the average silently.

Size the cluster from how the listener actually sits

Not from what decorrelates best — a spread wider than the seat optimises a place nobody’s head goes. On this sofa the honest envelope is **asymmetric**: sideways the head barely moves (ear separation is 17 cm, and you do not slide across the seat), but **front-to-back it moves ~ 20 cm** — slouching, and one or two cushions behind the back push you that far forward.

That is not a small effect. The 2026-04-28 sweeps were taken about 20 cm behind the 2026-08-10 ones — both perfectly normal seating — and below 200 Hz the two differ by **3 to 6 dB**, with the room’s upper-bass dominance moving from 4.1 dB to 2.8 dB. **A normal shift in how you sit changes the bass more than any panel move measured in this room, and more than the choice of house curve.** That is the argument for averaging: not statistical hygiene, but the fact that a single-point filter is tuned to a posture.

Recommended

positions	5 — the listening position, then ± 10 cm left/right and ± 20 cm forward/back, all at ear height. Asymmetric on purpose: it maps the seat, not a sphere
per channel	yes, L and R separately; do not average L with R at this stage
combine with	RMS average , after Align SPL ...
then	the averaged L and the averaged R become the L and R of the procedure — everything downstream is unchanged

If you stay single-point

Survivable, and this is the configuration the numbers in this guide come from. **Frequency-domain regularisation is the single-point substitute for spatial averaging** — both suppress narrow position-specific structure; the FDW just does it without moving the microphone. Accept these:

1. **The FDW is mandatory, not advisory.** With a spatial average you could argue for 20 cycles; with one point you cannot.
2. **Never boost a narrow null.** A deep narrow null is destructive interference between two arrivals — it is **not minimum phase**, so a minimum-phase inverse is the wrong *shape*, not merely the wrong size. The 0 dB max-gain setting enforces this; do not raise it.
3. **Keep the L/R average as the target** (step 5). It is already a partial average and it stops each channel chasing its own private nulls.
4. **Trust nothing below ~ 35 Hz.** The N801s are -6 dB around there anyway; correction buys excursion, not output.
5. **Above Schroeder (~ 166 Hz) single-point is fine** — the response there is direct sound and early reflections, which barely move over a seat.

The honest cost: a single-point correction is about as good as a multi-point one *at the microphone*, and noticeably worse one seat over. **It will not ring**, which is the thing you are fixing.

7. With several positions, do you still need the FDW?

Yes. Less of it, but you cannot drop it. This is the question people get wrong, so it is worth being precise about *why*.

Spatial averaging and time windowing answer **different** questions:

	spatial averaging	FDW
the question it answers	<i>is this feature real, or is it this seat?</i>	<i>how long may the correction be allowed to ring?</i>
removes	features that differ between positions	energy arriving late
what it does to a genuine high-Q room mode	nothing — it is at the same frequency everywhere	caps its Q at N

That second row is the whole argument. **A room mode survives spatial averaging with its Q intact** — that is what makes it a mode rather than an artefact. If the average still shows a $Q = 25$ peak at 45 Hz, it is real; and inverting it still builds a filter that rings

$$T60 = 2.2 \cdot Q / f_0 = 2.2 \times 25 / 45 = 1.2 \text{ seconds.}$$

Averaging established that the feature is genuine. It said nothing about whether correcting it is a good idea. **Only the FDW bounds the filter**, and §4's conservation law makes that inescapable: no number of microphone positions can make a 1 Hz-wide correction shorter than a second.

What multi-position buys you is permission to relax the FDW, because it is no longer doing two jobs at once:

measurement	FDW	reasoning
single point	12 cycles	must suppress interference <i>and</i> cap Q
5-position RMS average	15–20 cycles	interference already suppressed; only the Q cap is left
9+ positions, well spread	20 cycles	as above, with more confidence

Never off, and never above ~25 — beyond that you are no longer regularising anything and §5 becomes available again.

8. How much regularisation — choosing the FDW cycles

The **frequency-dependent window** is a Gaussian time window whose width varies inversely with frequency. Its width is set in **cycles**, and the conversion is exact and worth memorising:

A window of **N cycles** is N/f seconds wide at frequency f , so its bandwidth is f/N and its **fractional bandwidth is exactly $1/N$** . Therefore **N cycles $\leftrightarrow$ a cap of $Q = N$** .

FDW setting	fractional bandwidth	$\approx$ octave fraction
15 cycles (REW default)	1/15	$\sim 1/10$ octave — mild
12 cycles	1/12	$\sim 1/8$ octave — the starting point; see the N = 8 callout below
~ 9 cycles	1/9	$1/6$ octave
~ 4.3 cycles	1/4.3	$1/3$ octave

(a) the FDW is not one window but a *family*, one per frequency, of width N/f — at 12 cycles that is 480 ms at 25 Hz, 240 ms at 50 Hz, 60 ms at 200 Hz. (b) the same windows as frequency kernels: **equally wide on a log axis**, which is what constant fractional bandwidth means and why the FDW behaves like fractional-octave smoothing. (c) the real measurement through it at 4, 12 and 30 cycles.

The three constraints that pin N down

1. **The floor** — **do not truncate real modal decay**. N must exceed the Q of the modes you intend to keep. From the room's measured decay (R6):

at	credible T60	implied Q	cycles needed
50 Hz	~ 400 ms	9.1	9
80 Hz	~ 350 ms	12.7	13
120 Hz	~ 300 ms	16.4	16

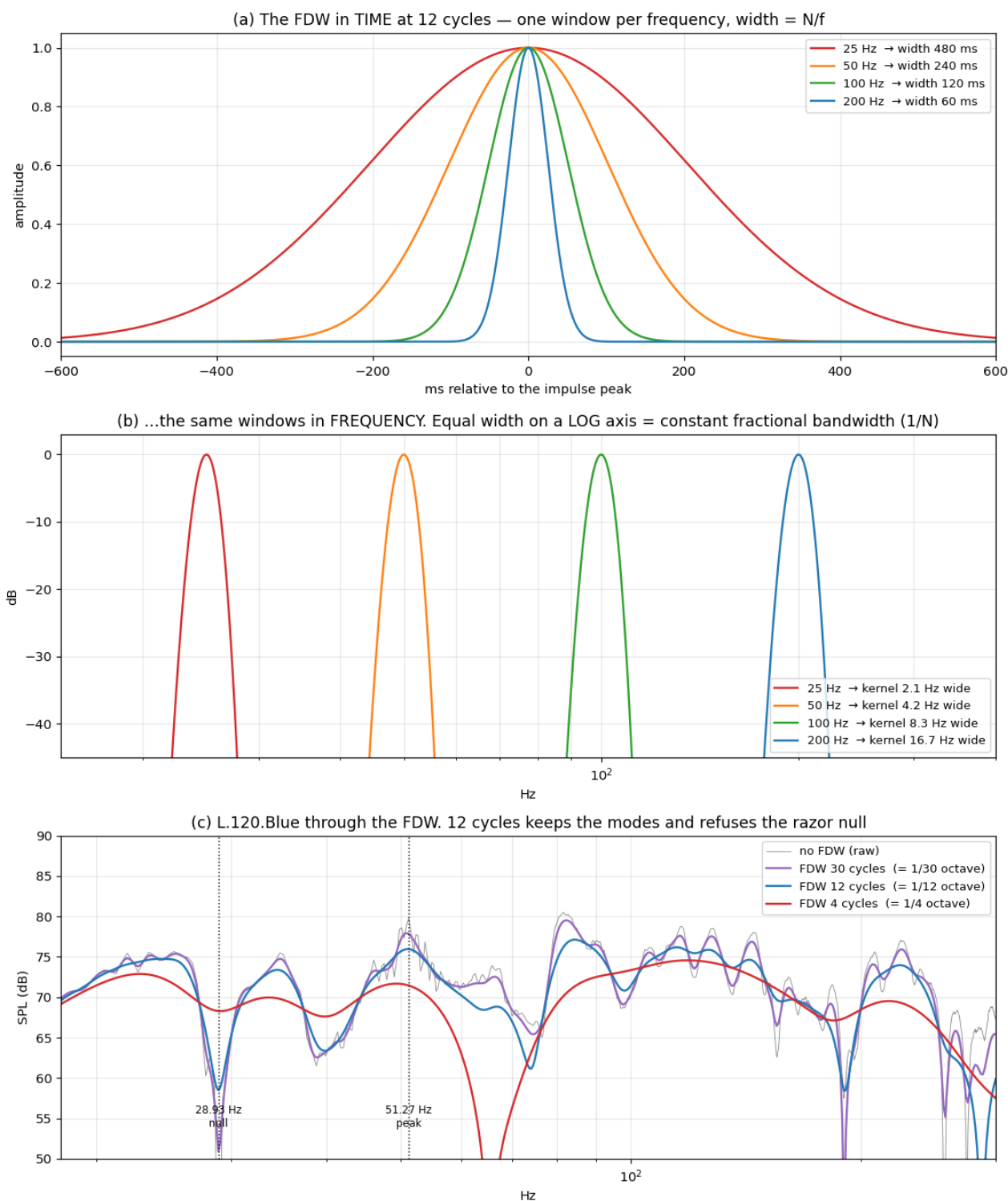


Figure 1: The FDW in time, in frequency, and applied to the measurement

→ $N \gtrsim 12$.

2. The ceiling — refuse position-specific interference. The features that wrecked the September filters had $Q = 40$ and $Q = 63$. N must be far below them.

→ $N \ll 40$.

3. The measurement must actually contain the window. At 25 Hz, $N = 30$ asks for 1200 ms of clean decay above the noise floor. It is rarely there.

$N = 12$ sits inside all three, and the floor and ceiling are a factor of 3 apart — this is not a knife-edge setting.

The floor is about modes. It does not apply to a cancellation — use $N = 8$

Constraint 1 says $N \gtrsim 12$ because N must exceed the **Q of the modes you intend to keep**. A speaker-boundary null has no Q to keep: it is two paths cancelling, not energy being stored and released. The floor simply does not bind on it, and where the sharpest feature in the band is a cancellation rather than a mode, **$N = 8$ is the better setting**.

Built both ways from the same 2026-08-10 measurements, everything else identical:

	FDW 12	FDW 8
74 Hz null, depth below its 60–95 Hz shoulders	−27.8 dB	−15.6 dB
narrowest feature, L / R	31.6 ✓ / 25.9 × bins	49.6 ✓ / 49.6 ✓
group delay, L / R	+24.1 / +24.5 ms	+17.2 / +17.4 ms
gated tones failing, L / R	2 / 5	1 / 0

The raw measurement puts that null 18.8 dB below its shoulders. **A 12-cycle window made it 27.8 dB — nine decibels deeper than it really is**, by stripping the late energy that fills it (the warning in “Two things the FDW is not”). The filter then built correspondingly steep shoulders, and those shoulders were the 63 and 79 Hz tails.

What it cost, measured as the change to the correction curve over 20–225 Hz: **0.58 dB rms on L, 0.77 on R**, and spent where you would want it — −1.4 dB at 74 Hz, +1.2 at 80, −1.3 at 100, and **0.0 to 0.3 dB below 63 Hz**. The §11 bass work is untouched and the delivered mono sum moved 0.11 dB or less in every band bar 160–225.

So the rule is not “12 is wrong” but: **N is chosen against the sharpest feature you intend to correct**. Identify what that feature is first. If it is a mode, constraint 1 applies and $N \gtrsim 12$. If it is a boundary cancellation — which this room’s is, at 74 Hz, predicted by geometry at 75.3 Hz — the floor is not in force and a shorter window is both more honest about the measurement and kinder to the filter.

What 12 cycles does to the feature that caused the trouble

The 38.3 dB razor null at 28.93 Hz:

	level at 28.93 Hz	depth below the 27.10 Hz peak
raw, no FDW	33.7 dB	38.3 dB
FDW 30 cycles	50.7	20.4
FDW 12 cycles	58.5	11.7
FDW 8 cycles	62.8	6.9
FDW 4 cycles	68.3	1.3

12 cycles turns a 38 dB canyon into a 12 dB dip — no longer something an inversion will build a Q-40 resonator to fill — while the surrounding modal structure survives intact.

Two things the FDW is not

It is not magnitude smoothing. It is a time gate, so it changes the **complex** response. Where late energy had been *filling* a direct-sound null, tightening the window can make that null **deeper**, not shallower — visible on the 4-cycle trace near 70 Hz in panel (c). The 28.93 Hz null shallows because it is late-arriving interference; a genuine early-arrival null would not.

It is not destructive. The window is a *derivation setting* over a retained impulse response: untick it, press Apply Windows, and the original response returns exactly. What is **not** reversible is anything you already derived from it — see step 2.

9. Which smoothing, and why the smoothing menu will not help you

Set the smoothing menu to whatever you find easiest to read. It will not reach the arithmetic. This is the single most important practical fact in the procedure, and it is documented:

*“The result of arithmetic on measurements that have compatible impulse responses is smoothed using the measurement A smoothing, **unsmoothed data is used during the calculations.**”* — REW help, All SPL Graph → Trace Arithmetic

That sentence reads as self-contradictory and is not. It joins two clauses about different objects: the **output** trace inherits A’s smoothing *setting* for display; the **input** to the math is the raw impulse response. Compute on raw data, then hand the result A’s display setting.

So the truthful summary is:

REW operation	uses smoothed data?
text / log export	yes — “the smoothing for the exported data”
EQ target match (Auto EQ)	yes — “apply ... before running the target match”
trace arithmetic on IR-compatible measurements	NO — unsmoothed, always
trace arithmetic on non-IR traces (e.g. 96 PPO imports)	yes, whatever they carried

Applying 1/6 octave to LX-MP and pressing divide **does nothing at all**. REW reaches past it to the impulse response.

What does reach the arithmetic is the window, because the window *defines* that impulse response. REW says so of the FDW specifically:

*“Any frequency-dependent settings are excluded, applying an FDW to the result would amount to applying the window twice, **as it is already applied to the data used to produce the result.**”*

That is why this procedure is built around the FDW and not around the smoothing menu.

If you ever do need smoothing to stick

Round-trip it: apply the smoothing, File → Export → Measurement as text **with that smoothing selected in the export dialog**, then re-import. The exported numbers carry the smoothing, and on re-import they *are* the data. This is redundant if the FDW is set, and is listed as optional in step 4a.

And do not use REW’s recommended smoothing here. “Variable smoothing is recommended for responses that are to be equalised” is advice for the **target-match / parametric** path, which carries its own filter-count, max-Q and max-boost guards. A bare inversion has none of them. Worse, Variable is **1/48 octave below 100 Hz**, which at these frequencies is not smoothing at all:

freq	1/48 oct width	in FFT bins (0.3662 Hz)
20 Hz	0.29 Hz	0.79
25 Hz	0.36 Hz	0.99
28.9 Hz	0.42 Hz	1.14
51.3 Hz	0.74 Hz	2.02
100 Hz	1.44 Hz	3.94

A kernel narrower than one bin is an identity operation. Below ~50 Hz, selecting 1/48 octave is identical to selecting None.

For *looking* at the response while you work, plain **1/6 octave** is the right choice — it shows you roughly what the FDW is doing to the data. Avoid **Psychoacoustic** for anything but viewing: it uses a cubic mean, a deliberately peak-biased estimator worth up to **+1.03 dB** on this dataset, which becomes over-cutting if it ever reaches an inversion.

10. The correction band

20 – 225 Hz.

edge	why
20 Hz low	below it the N801 is rolling off and the room measurement is mostly noise; correction there buys woofer excursion, not output above Schroeder (166 Hz) the response is direct sound plus early reflections — a property of the loudspeaker and the seat, not of the room. It is already good, it does not need a 128k-tap filter, and correcting it from one microphone point fits that point
225 Hz high	

The 225 Hz figure gives about half an octave of overlap above Schroeder, which is enough to let the correction die away gracefully rather than stop dead. REW blends the limits over **one octave**, so the correction is effectively fully out by ~320 Hz and fully in by ~160 Hz.

11. Below 80 Hz, correct the sum — not each channel

§5 warns against letting each channel chase its own nulls, and fixes it by giving both channels a **common target**. That is necessary and it is not sufficient. The divisor is still each channel’s **own** response, and below about 80 Hz that is enough to do real damage.

The failure this section exists to prevent

Measured on the 2026-08-10 Rscreen pair, at 50 Hz:

	L alone	R alone	RMS+phase average	vector average = the mono sum
level	75.92	71.83	73.19	68.45 dB

The two speakers **cancel each other** at the listening position across 45–56 Hz — the sum is 4.7 dB *below* the average of the parts. The RMS average cannot show this: it is a power average, it has no phase, and filling nulls is exactly what it is for.

So the divisor said “73.2 dB, target 71.4, cut it”, and the build cut **L by 4.20 dB and R by 0.00 dB**. But L was the *louder of two nearly anti-phase contributors*. Making them more equal made the cancellation more complete:

	before	after the per-channel build
mono sum at 50 Hz	68.45	65.96 dB
40–62.5 Hz, re midrange	–0.90	–3.49 dB

The region that was already the weakest in the response became the deepest hole in it, and the filter did that — correctly, by its own arithmetic.

The one-line reason

If both channels receive the **same** correction C, then **new sum = sum + C exactly**: the ratio between the two contributions never changes, so the depth of any cancellation between them is untouched. Only a **differential** correction can deepen a mono null.

Per-channel division against a common target is differential by construction. Above the transition it is what you want — it is how each speaker’s own response gets flattened. Below it, it is unforced error.

The fix is the divisor, not the target

The target does not change. Keep the EQ-window target built from the RMS+phase average exactly as §5 prescribes — a null in one path filled by the other, one common shape for both channels. Change only **what you divide by**, and only below 80 Hz:

Hz	target	cut asked vs RMS average	cut asked vs mono sum
20	69.79	4.98	5.05
25	72.21	3.99	4.27
31.5	72.56	1.34	0.62
40	72.17	0.00	0.00
45	71.81	0.00	0.00
50	71.36	1.65	0.00
56	70.89	2.78	0.00
63	70.20	2.73	2.40
71	69.53	4.27	3.55
80	68.84	7.19	5.17

Against the sum the filter asks for **nothing** at 45–56 Hz — not because anything was special-cased, but because the sum is already below target there and **the cut-only clamp engages by itself**. The deep-bass trim at 20–31.5 Hz survives intact. This is the whole technique: give the division a divisor that knows about the cancellation, and the existing guard does the rest.

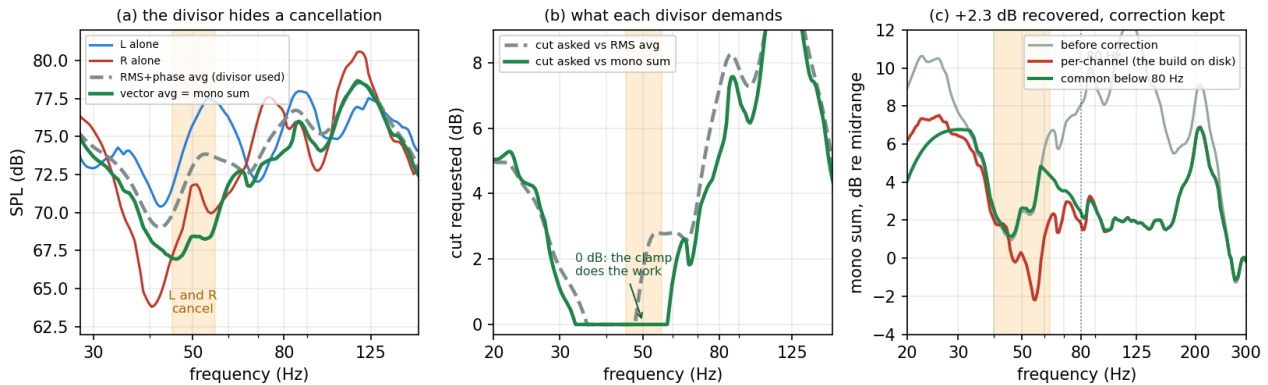


Figure 2: Why the divisor must be the sum below 80 Hz

Where 80 Hz comes from

Two independent arguments land on the same number.

Coherence. Below ~80 Hz the two speakers arrive at the seat correlated, so they sum as pressure and the sum is the physical quantity. Above it they decorrelate and the vector average stops meaning anything — measured here, RMS average minus vector average:

20	31.5	40	50	56	63	125	200 Hz
-0.07	+0.71	+1.23	+4.61	+4.55	+0.33	+0.16	-2.75

They agree within a decibel everywhere except the cancellation — and by 200 Hz the vector average has fallen 2.75 dB below, which is decorrelation, not information. **Use the sum only where it means something.**

How differential the correction is anyway. Channel minus average, rms:

	20–40	40–63	63–80	80–160	160–225 Hz
L	1.22	2.68	1.23	1.16	2.87
R	2.09	3.09	2.00	1.36	1.87

The differential is worst exactly in the cancellation band and modest elsewhere, so a crossover at 80 Hz removes the harm and forfeits almost nothing. This is also the standard bass-management crossover (Dolby, THX, ITU), chosen for the same reason: localisation cues are weak below it.

Building it: two divisions, spliced at 80 Hz

$$\text{FL} = \left[\begin{array}{c} \text{Target} \div \text{SUM} \\ \text{limited 20-80 Hz} \end{array} \right] \times \left[\begin{array}{c} \text{Target} \div \text{LX} \\ \text{limited 80-225 Hz} \end{array} \right]$$

Both factors are the same kind of object — **the target divided by a measurement** — and both are ordinary cut-only divisions. Outside its own limits each reverts to unity, so only one is ever doing work:

	below 80 Hz	above 80 Hz
$\text{Target} \div \text{SUM}$, limited 20–80	active	unity
$\text{Target} \div \text{LX}$, limited 80–225	unity	active
product	Target $\div$ SUM — common	Target $\div$ LX — per channel

The splice is seamless because REW’s two band-limit ramps are **complementary**: the roll-off at the upper limit of the first is exactly one minus the rise at the lower limit of the second, each a raised cosine over one octave. So in decibels the product is

$$\text{FL(dB)} = w \cdot (\text{Target} - \text{SUM}) + (1 - w) \cdot (\text{Target} - \text{LX}), \quad w: 1 \rightarrow 0 \text{ over } 57\text{--}113 \text{ Hz}$$

— a clean crossfade from the common correction to the per-channel one. Nothing steps, and at no frequency is any correction applied twice.

Both factors stay cut-only — nothing has to be allowed to boost

This is the property that makes the construction safe without a post-hoc check. $\text{Target} \div \text{LX}$ is a target-over-measurement division like any other, so **Maximum gain 0.0 dB on both factors**, and cut-only is then structural rather than something you verify afterwards.

The equivalent form, and why it is the worse way to build it

The same filter can be written as a common correction times a *differential* one, $(\text{Target} \div \text{SUM}) \times (\text{SUM} \div \text{LX})$, with the first factor spanning the full 20–225 Hz and the second limited to 80–225. SUM then cancels algebraically above 80 Hz and the result is the same filter — measured on the 2026-08-11 data, the two agree to **0.088 dB rms and 0.425 dB maximum** over 20–225 Hz.

Prefer the spliced form anyway. $\text{SUM} \div \text{LX}$ is a ratio of two *measurements*, so it genuinely needs to boost wherever a channel sits below the sum, and clamping it breaks the algebra:

	$(\text{T} \div \text{SUM}) \times (\text{SUM} \div \text{LX})$	$(\text{T} \div \text{SUM}) \times (\text{T} \div \text{LX})$
boost the second factor wants	+6.59 dB	+0.22 dB
Maximum gain it must be given	+6.0 dB	0.0 dB
maximum of the product, 20–225 Hz	+0.20 dB	+0.00 dB
cut-only is	verified after the fact	structural

The differential form ends up 0.20 dB *above* unity — harmless in itself, but it means the cut-only guarantee is no longer something the settings enforce. The spliced form never leaves it.

What it delivers

Predicted mono sum, dB relative to the untouched midrange:

band	before	per-channel build	common < 80 Hz	recovered
20–40	+8.06	+6.15	+5.82	−0.33
40–62.5	+2.68	+0.32	+2.63	+2.31
62.5–100	+7.85	+2.22	+2.89	+0.67
100–160	+10.25	+1.85	+1.85	0.00
160–225	+6.96	+4.38	+4.38	0.00

The 40–62.5 Hz loss comes back essentially in full, and the correction that was doing real work — the 100–160 Hz lump, 8.4 dB of it — is untouched.

This is not a local invention

Correcting the summed low-frequency field rather than each source is the mainstream approach: **bass management** in every multichannel standard sums below 80 Hz; **Dirac Live Bass Control** jointly optimises all bass sources rather than each independently; **Multi-Sub Optimizer** optimises the summed response at the seats for precisely the reason above; **Trinnov**, **Audyssey Sub EQ HT** and **Anthem ARC Genesis** all apply shared correction below the crossover; **Harman's Sound Field Management** optimises the combined field. The only unusual thing here is that the shared bass sources are the two main speakers rather than a subwoofer. The physics is identical.

If you cannot measure L+R

A measured L+R sweep is the honest divisor. Failing that, REW's **vector average** of **LX** and **RX** reconstructs it, and here that is trustworthy: the two measurements share one acoustic timing reference and their impulse peaks sit **0.013 ms** apart, which is 0.23° of phase at 50 Hz. Use the vector *average*, not a sum — it is level-consistent with the RMS+phase average the target was built from, so no 6 dB bookkeeping is needed.

Verify it when you can. A computed complex sum and a measured L+R have disagreed by 0.84–2.68 dB rms in this project before.

It has since been verified. The 185 cm configuration (DRC-185/L0.txt, R0.txt, LR.txt) includes a genuine measured L+R sweep alongside the two channels. Computing the complex sum of L0 and R0 reproduces the measured L+R to **0.50 dB rms over 20–250 Hz**. The substitution is sound.

12. What a magnitude filter cannot do — and the one cheap way round it

Everything above corrects **magnitude**. There is one low-frequency failure it cannot address at all, and one inexpensive tool that can. This section is the worked case, from the 185 cm configuration in this repository.

The failure

Speakers 1.85 m from the front wall, microphone at 4.54 m, measured L+R:

deepest point of the mono sum, 20–300 Hz	– 17.9 dB re midrange, at 41 Hz
L alone, at its own worst	–8.3 dB
R alone, at its own worst	–10.0 dB

The sum is far worse than either channel, and it has two stacked causes:

	each channel already down	L and R cancel a further	total
36 Hz	–5.6 dB	–4.5 dB	–9.8
40 Hz	–6.0 dB	–7.3 dB	–13.1
43 Hz	–5.6 dB	–5.7 dB	–11.1
46 Hz	–3.7 dB	–3.5 dB	–7.0

The first column is **SBIR**: 1.85 m from the front wall gives an excess path of 3.70 m and a null at $343/(4 \times 1.85) = \mathbf{46.4\ Hz}$, and it hits both channels identically because both sit at the same distance. The second is that L and R arrive at the seat **142° apart** at 40 Hz and partially cancel.

Neither is reachable by a cut-only filter, and boosting is not a serious option: +11.7 dB at 41 Hz is **15× electrical power** and **3.9× cone excursion**, spent on a null that moves with the listener's head.

The half of it that is free

The SBIR component is geometry — nothing but moving the speakers fixes it, and that is what the 120 cm configuration is (it puts the null at 71.5 Hz).

But **the 7.3 dB of L/R cancellation is not a magnitude problem at all**. It is two sources at 142° , and rotating one of them costs no energy, no headroom and no excursion. A single second-order **allpass** does it. Measured — L0 through a 42.5 Hz, Q 2.5 allpass, summed with R0:

band	L + R	L_ap + R	change
20–32 Hz	+1.5	+1.6	+0.1
32–50 Hz	−11.8	−4.5	+7.3
50–63 Hz	+3.2	+3.2	−0.0
63–100 Hz	+6.0	+5.5	−0.5
100–225 Hz	+0.6	+0.5	−0.0

Deepest point **−17.9** → **−7.3 dB**; at 43 Hz the gain is **+11.7 dB**. Seven decibels recovered in the problem band, the rest of the range moved by less than half a decibel, and **not one decibel of boost applied anywhere**.

What it costs, and what it makes the filter

The measured allpass: magnitude flat to **0.26 dB** across 20–500 Hz, peak group delay **37.3 ms at 41.7 Hz**, falling to 1.9 ms by 80 Hz.

That 37 ms sits right at the low-frequency group-delay audibility threshold, and it is applied to one channel only. Below 80 Hz that is benign — the same reason bass management crosses there: localisation cues are weak, so an L/R group-delay mismatch has little to act on.

This makes the filter mixed phase — but the harmless kind

Any causal stable filter factors as *minimum phase* × *allpass*, and is minimum phase exactly when that allpass factor is trivial. Adding one makes it non-trivial, so §2’s “we restrict ourselves to minimum phase” no longer describes the chain.

It is still safe, because there are two different operations both called phase correction:

	what it does	causality	cost
add an allpass	re-times one source against another	causal	group delay only
invert the room’s excess phase	undoes an allpass the room applied	acausal	pre-ringing

The second is acausal by necessity — the inverse of a stable allpass has its poles outside the unit circle, so no causal stable inverse exists. That is the operation that needs windowing and that pre-rings.

This filter is the first kind. Its impulse response was checked: **16384 taps, peak at sample 0, exactly zero energy before the peak**. It can only delay. There is nothing in front of the impulse to pre-ring with.

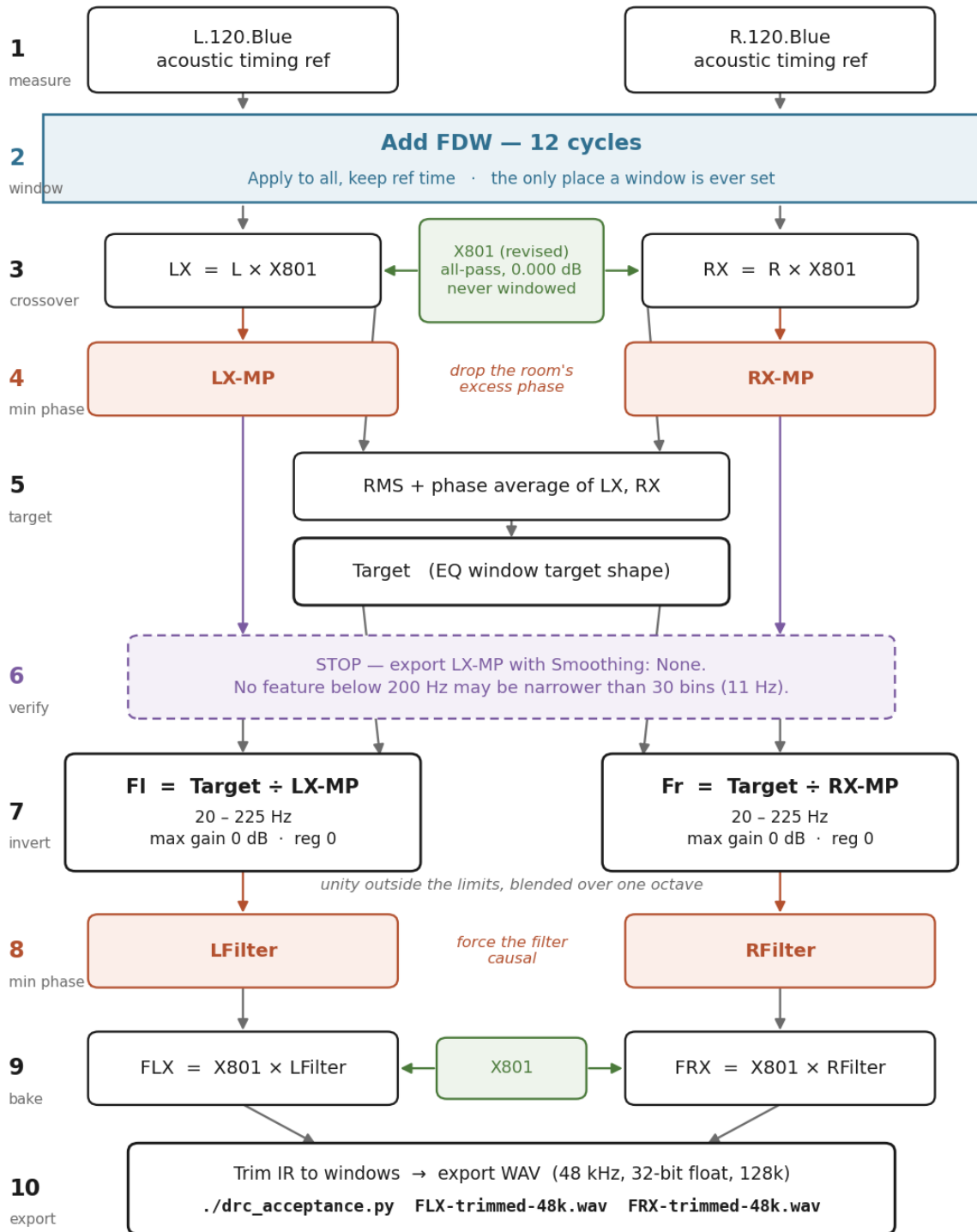
What this is, in the wider landscape

Plain two-channel **Dirac Live would not find this**. It measures L and R separately and corrects each toward the target — which at 142° is exactly the operation §11 warns about, driving the cancellation *deeper*. The tools that find it are the ones that optimise the **summed** field across sources: **Multi-Sub Optimizer** (free, and the closest match to what was done here), **Dirac Live Bass Control / ART**, and **Trinnov** source optimisation.

REW can generate the filter natively: the EQ window offers an **All-Pass** filter type taking frequency and Q, and it exports into the impulse response like any other. No mixed-phase design software is required.

When to reach for it

Only when the mono sum is materially worse than either channel alone. Test it in one step: compare **L+R** against **max(L, R)**. Coherent summation gives about +6 dB; anything at or below 0 dB means the channels are fighting, and that deficit — not the magnitude response — is what to fix first. Above ~80 Hz leave it alone: the phase relationship there varies too fast with position to be worth chasing, and localisation does depend on it.



Every name is a real trace in ../DRC-120.blue/120.blue-with-inversion.txts/ ; the operations are the ones recorded in their export headers.

Figure 3: The REW inversion chain

Part III — The procedure

Eleven steps (the diagram folds 10 and 11 into one box). Every trace name below is a real trace in `../DRC-120.blue/120.blue-with-inversion.txts/`, and every operation is the one recorded in that file's export header — this is the chain that was actually run, with the guards added.

Step 1 — Measure L and R

Do:

setting	value
sweep	256k log swept sine , one sweep
level	-12 dBFS
timing	acoustic timing reference — required
sample rate	48 kHz
channels	L and R separately , same microphone position
name them	L.120.Blue, R.120.Blue

Multi-position: sweep all 5 positions per channel, **Align SPL...**, then **RMS average** (§6). The RMS averages then *are* L and R for everything below.

Why the acoustic timing reference: it puts $t = 0$ at the impulse peak. The FDW is centred on the window reference time, and REW's help is explicit that “for best results this should be at the peak of the impulse”. Without a timing reference the reference time is an estimate and the window sits in the wrong place — quietly, with no error message.

Step 2 — Set the window. Once. Here. Nowhere else.

Do: open **IR Windows** on L. Change **one** field:

control	set to
Left window shape / width	leave (Rectangular, 500 ms)
Right window shape / width	leave (Rectangular, 1000 ms)
Ref Time	leave — it is the IR peak
Add FDW	✓ ticked, 12 cycles

Then press **Apply to all, keep ref time**.

Apply it to **L and R only**. Never to X801. Never again downstream.

Why only the FDW: at 12 cycles the FDW is narrower than the 1 s right window everywhere above 12 Hz, so it governs the entire working band on its own. The 1 s rectangular truncation is harmless because the decay is already in the noise by then. Shortening the rectangular right window instead would manufacture ripple (R2); this route avoids the question entirely.

Why keep ref time and not Apply windows to all: each measurement has its own reference at its own IR peak. Plain “apply to all” overwrites those with *this* measurement's Ref Time and misaligns the others. “Keep ref time” copies the shapes and widths and leaves each measurement's own peak alignment intact.

L and R showing different Ref Times is correct, not a fault

Every measurement carries its own Ref Time because every sweep began at a different instant in its own capture. What matters is whether the IRs are aligned *after* referencing, and with an acoustic timing reference they are: reconstructing the IRs from the exported 2026-08-10 sweeps puts every peak within **0.013 ms (4.5 mm)** of its own $t = 0$, and the L–R offset is **4.5 mm** — the two speakers are equidistant to the millimetre, exactly as the geometry says.

And this procedure is immune to it regardless. The target is built with an **RMS average**, which is magnitude-only, and the divisor is taken to **minimum phase** in step 6, which discards

measured phase altogether. No step downstream of the window can see a timing offset between L and R. Timing would only matter for a **complex (vector) L+R sum** — which this guide never asks for, and which §2's caveat 1 warns against for its own reasons.

Why this is the only window you will ever set — three quotes from the Trace Arithmetic documentation:

- “The currently applied impulse response window settings are used for each trace.” → what you set on L and R is what every downstream operation consumes.
- “The result uses the same window settings as trace A” → derived traces inherit automatically. LX = L × X801 is built with L as trace A, so LX already carries L's window. There is nothing to re-apply.
- “Any frequency-dependent settings are excluded, applying an FDW to the result would amount to applying the window twice.” → re-applying is not neutral, it is a second regularisation. REW deliberately clears the FDW flag on results. **Do not put it back.**

And one thing you cannot repair afterwards:

“...or for division and inversion, which use windows that span the entire resulting impulse response as the result is typically not causal.”

The divide's output is deliberately un-windowed. **If the inputs were wrong, the output cannot be fixed.** This is why step 2 comes before everything.

[!] **The window is reversible. The traces built from it are not.**

Untick the FDW, press Apply Windows, and L returns to exactly what it was. But trace arithmetic creates a **new measurement, frozen at the moment you press the button**. Change the window on L afterwards and LX, LX-MP, the division and the exported WAV all keep the old data — they do not recompute.

Set the window first, then do the arithmetic. If you change the window, redo the chain from step 3.

Step 3 — Bake the crossover correction into the measurement

Do:

1. File → Import → Impulse Response → X801.wav. Name it X801 (revised). **Leave every window control on it alone** — see the warning below.
2. Trace Arithmetic: LX = A × B, A = L.120.Blue, B = X801 (revised).
3. Likewise RX = R.120.Blue × X801 (revised).

Why: you are going to invert the system *as it will actually play*, and it will play through the crossover correction. Everything you then look at on screen — LX, RX and their average — is the real corrected system.

A useful thing to know: because X801 has magnitude 0.00000 dB everywhere, |LX| = |L| exactly, and therefore LX-MP and L-MP are **identical**. Step 3 is magnitude-neutral, and the final filter would come out the same if you skipped it. Do it anyway, for two reasons: the traces you inspect are then honest about what you will hear, and if X801 is ever replaced by something that is *not* a pure all-pass (Xo801, which adds a bass-alignment term, is such a thing) the step becomes load-bearing without warning.

[!] **Never window X801**

It is an all-pass whose energy is spread symmetrically over ±1365 ms by design. An FDW or a shortened right window would truncate the very phase rotation it exists to apply, silently degrading the crossover correction.

Note that inheritance is correct here without you doing anything: LX is built with A = L, so it inherits L's FDW ✓. In step 9, FLX is built with A = X801, so it inherits X801's window — i.e. none ✓, which is right, because a finished filter should not be windowed.

Step 4 — Minimum phase, first time

Do: on LX, **Generate minimum phase** (same control panel as Trim IR to windows). Name it LX-MP. Likewise RX-MP.

dialog option	set to	why
Cal file effects	included	you are modelling the acoustic response as measured, and the mic calibration is part of what the measurement means
LF tail	yes, corner at 15 Hz	required — without it the minimum-phase transform corrupts the magnitude it is supposed to preserve. See the callout below
HF tail	no	measured: the error above 1 kHz is 0.000–0.002 dB. The traces already run to 24 kHz = Nyquist, so there is no top-end edge to ring against

[!] The LF tail is not optional — this guide said “no” and was wrong

A minimum-phase copy has exactly one obligation: **leave |H| unchanged**, and supply the phase the magnitude implies. With **No LF tail** REW does not satisfy it. Measured on the 2026-08-12 build, max |error| in dB, where every entry should be zero:

conversion	5–15 Hz	15–25 Hz	25–40 Hz	40–225 Hz
LX → LX-MP	13.41	0.58	0.03	0.33
RX → RX-MP	8.88	0.53	0.03	0.04
SUM → SUM-MP	11.15	0.56	0.02	0.03
FL → LFilter	0.21	2.41	0.30	0.13

The mechanism. Minimum phase is obtained by a Hilbert transform of the log magnitude, which is a *global* operation — it needs the response defined across the whole spectrum. Truncate it and the transform rings against the edge, decaying away from it. That is why the error is largest at the bottom of the data and falls to ~0.03 dB by 25 Hz, and why the top end is spotless: the traces run to Nyquist, so there is no upper edge at all.

Why the corner goes at 15 Hz, not the default 20. Real sweep data starts at **15.01 Hz**; everything below is already synthetic, inherited from X801’s wider span through the trace multiply. A corner at 15 Hz replaces only synthetic data. A corner at 20 Hz would extrapolate over 5 Hz of genuine measurement — and that sits inside the band-limit blend (14.1–28.3 Hz), so it reaches the filter. If the field will not go below 20, take 20; the cost is small because the filter is fading toward unity through there anyway.

Put the corner at or below the first measured point, not just above it. The 2026-08-12 rebuild used 16 Hz against data starting at 15.0146 Hz, and the ~1 Hz of overlap left a **single-bin –140 dB null at exactly 15.01 Hz** in the exported WAV, where the tail splices onto the measurement. It is harmless — one bin, below the correction band, in a region already 29 dB down, and R8 does not see it — but it is free to avoid. Set the corner to the sweep’s start frequency and the splice has nothing to disagree with.

Slope: make the tail resemble what the trace physically is. For a measurement, the speaker’s own roll-off — **24 dB/oct** for a ported box, 12 for a sealed one.

Verify, do not assume. Export the copy and its source with **Smoothing: None** and subtract. $|LX-MP| - |LX|$ must sit at the ~0.03 dB level across the whole correction band. This is a two-minute check that catches a class of error nothing downstream can undo.

Why this step exists — it is the most important conceptual step in the chain. You are about to divide by this trace. What you divide by determines what gets inverted:

divisor	what the filter would try to undo	verdict
LX (the raw measurement)	magnitude and the room's excess phase — every reflection, at one point	acausal, pre-rings, valid at one microphone position
LX-MP	magnitude, and exactly the phase that the magnitude implies	✓ causal, minimum delay, robust

Dividing by the raw measurement is the single most common way to produce a filter that measures beautifully at the microphone and sounds wrong. The minimum-phase copy throws away the part of the response that is not correctable, before it can reach the division.

Step 5 — Build the target

Do:

1. Select LX and RX → **RMS + phase average**. Result: **RMS + phase average**.
2. Open it in the **EQ window** and save its shape as the target: **Target RMS + phase average**. Adjust the house curve here if you want one.

Why the L/R average and not each channel's own response: it is already a partial spatial average — a null in the left path is filled by the right — and it gives both channels a **common** target, so correction does not pull the stereo image. Targeting each channel at its own smoothed response instead would let each chase its own private nulls, which is exactly the failure in §5.

Why go through the EQ window: what comes out is a **smooth curve**, not a copy of the measurement. It is also where the house curve goes. Note that a target shape is **magnitude only**, which is fine: only its magnitude survives step 8.

The target level — let REW calculate it, then check it this way The house curve is a **shape**, defined as 0 dB at 200 Hz. The **target level** is what anchors that shape in absolute SPL, and it is the single number that decides whether the corrected bass joins the rest of the spectrum or sits above or below it.

Press Calculate and let REW set it. REW anchors the target to the speaker's **midrange reference level**, which is the right choice and is not obvious. Measured on the 2026-08-10 LR.rms+phavg:

region	level
correction band, 20–225 Hz	74.07 dB
300 Hz – 3 kHz (midrange)	68.80 dB
REW's calculated target level	69.20 dB

REW's number lands within **0.4 dB of the midrange**, and the correction band runs **5.3 dB hot** against it. That elevation is exactly what the filter is there to remove.

The check that matters: does the band edge join what is above it?

The filter is unity above 225 Hz, so whatever the correction leaves at the top of the band must match the untouched response just above it — otherwise you build a step at 225 Hz. With the calculated level:

180 Hz	200 Hz	225 Hz	225–315 Hz (untouched)
68.4	68.7	68.9 dB	68.5 dB

Continuous to a few tenths. **Do not** set the target level by matching the measurement's own in-band level at 225 Hz — that keeps the room's bass elevation instead of removing it, and leaves the whole band ~4 dB above the midrange. If REW's calculated level looks “too low”, this is why: the number to compare it against is the **midrange**, not the neighbouring bass.

The target’s LF cutoff — move it to 5–10 Hz, below the correction band The target shape carries a **low-frequency cutoff**, and REW’s default puts it at **20 Hz, 24 dB/oct** — right at the bottom edge of the match range. Move it to 5 or 10 Hz.

Does that corrupt the calculated target level, given the speakers produce nothing at 10 Hz? No — and the reason is worth being clear about, because the worry is reasonable. REW fits the target to the measurement over the **match range**, 20–225 Hz. What the target does *below* 20 Hz is outside the fit entirely and never enters the calculation. What a 20 Hz cutoff does instead is bend the target **inside** the match range:

target attenuation from the LF cutoff	20 Hz	22 Hz	25 Hz	31.5 Hz	40 Hz
cutoff at 20 Hz	–3.01	–1.66	–0.67	–0.11	–0.02 dB
cutoff at 10 Hz	–0.02	–0.01	–0.00	–0.00	–0.00 dB
cutoff at 5 Hz	–0.00	–0.00	–0.00	–0.00	–0.00 dB

With the cutoff at 20 Hz, **10.9 % of the match range is pulled down by more than 0.5 dB**; at 10 Hz or 5 Hz, none of it is. Averaged across 20–225 Hz the depression is **–0.185 dB** at $f_c = 20$ and **–0.001 dB** at $f_c = 10$, so moving the cutoff shifts REW’s calculated target level by about **0.2 dB** — the target no longer has to sit slightly high to compensate for its own bend. That is the whole effect on the level, and it is negligible.

It does not make the filter reach below 20 Hz either. The division is band-limited to 20–225 Hz and reverts to unity below, blended over one octave; and max gain 0 dB clamps every boost regardless. Extending the cutoff changes **the target’s shape, not the filter’s reach** — there is no excursion risk from a target that is defined at 10 Hz.

Why this should clear two of the three acceptance-test failures

On the 2026-08-11 build, two of the three [R8](#) failures localise to **20.32 Hz** (narrowest feature) and **19.59 Hz** (group delay) — both sitting on the band edge, not on anything acoustic.

A 24 dB/oct corner *inside* the correction band is a steep feature, and the division reproduces it faithfully in the filter. Steep means high Q, and high Q means simultaneously a narrow feature and a long group delay — which is precisely the pair of tests that fail, at precisely the corner frequency. It is the same trade as §4: you cannot have the sharp edge and the short filter.

Putting the corner at 5–10 Hz leaves the target smooth across the whole 20–225 Hz band, so the filter has no steep band-edge feature to build and both tests should clear. The third failure is a separate matter and this will not touch it. **Verify on the rebuild rather than assuming** — this is a prediction from the mechanism, not a measured result.

Choosing a target the filter can actually reach This is the decision that determines whether the correction does anything at all, so it is worth more than a moment. With max gain at 0 dB the filter only cuts, therefore:

achieved = min(target, measurement)

A target that sits **above** the measurement produces no correction whatever — not a partial one, none. So the target is not a wish. It is a **ceiling you slide down under the measured curve**, and the measurement is its upper bound.

Measured here after step 2 (FDW 12 cycles), in dB relative to each channel’s own level at 200 Hz — i.e. exactly that upper bound:

	20	25	31.5	40	50	63	80	100	125	160	200 Hz
L	+3.7	+6.1	+1.9	–5.2	+6.9	+0.1	+6.0	+2.7	+7.2	+1.0	–0.1
R	+4.6	+7.8	+2.8	–0.8	+2.0	+5.0	+11.2	+0.3	+9.4	+6.5	0.0

This room is not bass-shy at 120 cm. Below 160 Hz it runs *above* its own 200 Hz level almost everywhere. What it actually has is **lumps at 80 and 125 Hz** and a **hole near 40 Hz** — and the hole is 4.4 dB deeper on L than on R, so it is one channel’s path, not a room property, and must not be filled (§6 rule 2). The job here is mostly **cutting**, not lifting.

[!] **Ask for more lift than the room offers and you get no correction at all**

The deployed chain used a **+6.5 dB** shelf. Against the table above that target sits over the measurement across most of the low band, so the clamp engaged and the filter did nothing:

	25 Hz	35 Hz	45 Hz	63 Hz
correction produced	0.00 dB	+0.01	+0.01	+0.01

Roughly half the correction band received no correction. A house curve you cannot reach is not a gentle preference — it silently switches the filter off.

Four candidate shapes, and what each actually delivers Computed as $\min(\text{target}, \text{measurement})$ on the real FDW-12 data, both channels, everything in dB relative to 200 Hz:

target	deep 20–45	upper 63–160	balance	ripple	band cut	deepest cut
<i>no correction (the room)</i>	+2.8	+6.1	−3.3	4.21	—	—
A as deployed, +6.5 shelf	+2.6	+1.6	+1.0	2.59	57 %	8.6 dB
B reachable, +3.0 shelf	+1.1	+0.3	+0.8	1.84	68 %	10.0 dB
C +4 low, −3 upper-bass	+1.5	−2.2	+3.7	2.07	78 %	12.9 dB
D flat	−0.8	−0.2	−0.6	1.41	74 %	10.6 dB

“**Balance**” is the number that decides how much bass you hear: deep minus upper, i.e. how far the bottom two octaves stand clear of the upper bass. Read the first row and the problem is obvious — **this room is upper-bass dominant by 3.3 dB**. That is the 80 and 125 Hz lumps, and they mask everything below them.

- **D (flat) is the wrong answer.** It cuts the deep bass along with everything else and ends up with *less* low-end weight than doing nothing.
- **A (as deployed) is doing least of all** — 57 % of the band, because most of the shelf is unreachable and clamps.
- **C is the one to use if you want low end.** It gives **+3.7 dB of balance against A’s +1.0** — nearly 3 dB more apparent weight — with *less* ripple than A, and it costs only 1.1 dB of absolute deep bass relative to A.

More bass comes from a deeper cut, not a taller shelf

With a cut-only filter a taller shelf means *less* cutting, so it buys loudness and lumpiness together. The lever that actually delivers low end is the **upper-bass scoop**: pulling 80–160 Hz down un-masks the 20–45 Hz region that is already there. Cutting is always reachable, so this direction is free in a way that lifting never is.

Recommended: shape C — about **+4 dB below ~45 Hz**, through 0 dB near 70 Hz, to a broad **−2 to −3 dB centred around 110–120 Hz**, back to 0 dB by 200 Hz. Start the scoop at −2 dB and deepen it by ear; too much hollows out male voice and cello. The extra cut costs headroom (12.9 dB at the worst peak, against 8.6 dB now), which is amplifier gain, not excursion.

Two things the target cannot do:

- **Anything above 225 Hz.** The filter is unity there (step 7), so the target’s shape above the band is decoration. If you want a treble tilt it has to be a separate broad shelving filter, not this target.
- **Fill the 40 Hz hole.** It is destructive interference on one channel. Boosting it costs excursion and does not fill it.

(If you want more low-end than the reachable ceiling allows, the honest fixes are placement and treatment, not the target — the same conclusion as §5’s.)

Step 6 — Stop and verify

Do: export LX-MP with **Smoothing: None** and run **two** checks.

6a. The minimum-phase copy must have preserved the magnitude. Export LX too, subtract, and require $|LX-MP| - |LX|$ to sit at the **~0.03 dB** level across 20–225 Hz. Repeat for RX-MP and SUM-MP.

This is a property, not a tolerance: a minimum-phase copy changes phase and nothing else, so any visible deviation is an artifact. If it fails, the LF tail in step 4 is off — see the callout there. Nothing downstream can undo an error in the divisor, because everything downstream divides by it.

6b. No *peak* below 200 Hz may be both narrower than ~30 FFT bins (≈ 11 Hz) and more than ~6 dB above its surroundings.

Why peaks and not dips. Because max gain 0 dB (step 7) already disarms the dips, and only the peaks reach the WAV:

in the divisor	what the division makes of it	survives?
narrow dip	filter wants a narrow <i>boost</i> → clamped to unity	no
narrow peak	filter builds a narrow <i>deep cut</i>	yes

This matters, because an honest divisor routinely contains narrow dips and they are not a defect. On the 2026-08-11 build:

trace, un-smoothed, 20–200 Hz	narrowest peak	narrowest dip
LX-MP (the divisor)	14 bins, 3.9 dB @ 35.5 Hz	6 bins, 40.1 dB @ 188 Hz
RX-MP (the divisor)	23 bins, 4.1 dB @ 50.5 Hz	13 bins, 4.1 dB @ 57.9 Hz
F1 (what ships)	38 bins, 5.8 dB @ 98.9 Hz	12 bins, 3.0 dB @ 35.5 Hz

A rule applied to *any* feature would have rejected LX-MP over a 6-bin dip and thrown away a build whose filter is comfortably inside the threshold. The same applies to the 74 Hz front-wall null that a 12-cycle FDW exposes on the left channel (§8): across 60–90 Hz LX-MP swings **35.7 dB**, while F1 swings **11.7 dB** — at the null itself the filter sits at exactly 0.00 dB, unity, with the cut confined to the shoulders. **Windowing revealing a null is working as intended; it is not a Step 6 failure.**

Why the FDW still matters even so. The clamp defuses dips, not peaks, and un-windowed modal peaks are exaggerated by reverberant energy that belongs to the room and not to the direct sound — invert those and you cut deep narrow notches fitted to one microphone point. It also keeps the shoulders around a null from being over-cut right next to a unity notch. §5’s disaster was a dip, and cut-only alone would have blunted it; the FDW is what keeps the rest of the filter honest.

If a narrow tall peak does survive, the FDW in step 2 did not take. Common causes: it was applied to the wrong measurement; Apply Windows was never pressed; or the arithmetic in step 3 was done *before* step 2 and is holding stale data. Thirty bins is the acceptance threshold from R8, applied one step early — and the surest version of this test is to run it on F1 itself once step 7 is done.

(Optional belt-and-braces, step 4a: *bake the smoothing in by round trip* — apply 1/6 octave to LX/RX, export as text **with that smoothing selected in the export dialog**, re-import, and take the minimum phase of the re-imported traces. Redundant if step 2 was set properly. Doing neither is what produced §5.)

Step 7 — The division

Two divisions and a multiply, not one division. One shared setup (7a, 7b) and then two operations per channel (7c, 7d). The reason is §11: below 80 Hz the two speakers cancel each other at the seat, and dividing by each channel separately deepens that cancellation.

7a — the sum, and its minimum-phase copy. Select LX and RX → **Vector average** → name it SUM. (If you have a measured L+R sweep, use L+R × X801 instead — better. See §11’s last note.) Then **Minimum phase copy** → SUM-MP.

7b — the common filter, below 80 Hz. Trace Arithmetic, **A over B**:

field	value
A	Target RMS + phase average
B	SUM-MP
Lower / upper frequency limit	20 Hz / 80 Hz
Maximum gain	0.0 dB
Regularisation	leave at 0 — see below

Name it **Fcommon**. There is only one, shared by both channels.

7c — the per-channel filters, above 80 Hz. Trace Arithmetic, **A over B**:

field	value
A	Target RMS + phase average — <i>the same A as 7b</i>
B	LX-MP (the minimum-phase copy, not LX)
Lower / upper frequency limit	80 Hz / 225 Hz
Maximum gain	0.0 dB
Regularisation	0

Name it **Fper_L**. Repeat with B = RX-MP for **Fper_R**.

7d — combine. Trace Arithmetic, **A times B**: **Fcommon** × **Fper_L** → **F1**. Same with **Fper_R** → **Fr**.

Both factors are cut-only, so the product is cut-only: **no maximum-gain setting anywhere in this step is above 0.0 dB, and none needs to be.** The 80 Hz splice is seamless because REW's band-limit ramps are complementary — §11 has the algebra and the measured agreement.

(To reproduce the older single-division behaviour — per-channel everywhere — skip 7a and 7c, and in 7b use B = LX-MP with limits 20 Hz / 225 Hz. The tables below were measured on that version, and the boost/cut asymmetry they show is unchanged.)

Why the frequency limits are what makes this a filter, not a curve. Outside the limits the result reverts to **unity gain**, blended over one octave centred on the limit. Measured on the deployed file:

10 Hz	20 Hz	225 Hz	250 Hz	300 Hz	1 kHz	15 kHz
+0.04 dB	+0.02	−3.26	−0.25	+0.01	+0.01	+0.01

Unity to within 0.04 dB outside the band, and the one-octave blend is visible in the 225 → 250 → 300 Hz run. Without the limits you would be exporting the whole target curve as a filter.

What max gain 0 dB actually does: the filter only ever cuts

Read off the deployed chain — the correction the target asked for, against what the division produced:

	25 Hz	35 Hz	45 Hz	63 Hz	80 Hz	125 Hz
Target — LX-MP (wanted)	+2.2	+4.3	+2.6	+2.5	−2.0	−4.4 dB
what came out	0.00	+0.01	+0.01	+0.01	−3.7	−4.8 dB

Every boost is clamped flat; every cut passes. That is the setting doing exactly what it says, and it is the right conservative choice — a deep null at one microphone point is destructive interference, not a deficiency of output, and filling it with gain wastes excursion and does not fill it.

Know what it implies: **the correction is cut-only.** It will lower the peaks toward the target and leave the dips where they are. If your target looks nothing like the result, this is why.

(Figures read off the 96 ppo exports, which carry display smoothing; the divide itself ran on unsmoothed data, so treat the decibels as indicative and the boost/cut asymmetry as exact.)

Why regularisation is left at zero here, and when it would matter. Regularisation puts a floor under the divisor so the quotient cannot run away where the divisor is tiny. It is the right tool for one specific disease: **a deep null being filled with gain.** With **max gain at 0 dB** there is **no gain anywhere**, so it has nothing to act on — and this is not an argument from principle, it is what the deployed filter measures:

FLX-trimmed-48k.wav, 20 – 225 Hz	
maximum	+1.19 dB
minimum	–7.24 dB

The filter never boosts. A boost limiter cannot change it.

Set regularisation only if you raise max gain above 0 dB to allow broad dips to be partly filled. Then choose it by the readout, not by a number from a guide — REW “*shows the maximum gain corresponding to the regularisation percentage next to the control*”, so dial the percentage until that reads the gain you intend. For orientation, the depths that exist in this data, measured relative to the divisor’s own average level over 20–225 Hz:

divisor	deepest dip below its average
L, raw (no FDW)	–30.4 dB
L, FDW 20 cycles	–24.1 dB
L, FDW 12 cycles	–13.7 dB
R, FDW 12 cycles	–8.5 dB

After step 2 there is no canyon left to regularise — **the FDW has already done that job.** Note the trap in these numbers: the small percentages are the ones that reach a *raw* razor null, precisely because a razor null is deep. A setting of 3 % puts the floor around 30 dB below average, which is 16 dB below anything that survives step 2. Small numbers here are not gentle; they are inert.

[!] **The feature that caused 1.3 seconds of ringing is 3.5 dB deep**

FLX-trimmed-48k.wav at native 0.3662 Hz resolution, around the null. These are absolute filter gains, not relative to anything:

28.198 Hz	28.564	28.931	29.297
+0.87 dB	–1.64	–2.35 dB	+1.19
GD +71 ms	–101	–105 ms	+81

The bottom of the dip sits at **–2.35 dB**. What matters is the **excursion across it**: from +1.19 dB in the bin next door down to –2.35 dB is **3.54 dB**, and it happens over **two bins** — **0.73 Hz**. That is the whole feature. It carries **105 ms of group delay** and rings for **1348 ms**.

Depth is not what hurts. Width is. A 3.5 dB wiggle you would never notice on an amplitude plot is a Q-40 resonator because $Q = 28.93 \div 0.73 \approx 40$, and §4’s law converts 0.73 Hz directly into 1.4 seconds. This is the whole reason step 2 exists, and the reason no guard in this step can substitute for it.

(How that dip got there, since the clamp had pinned the magnitude flat: *LX-MP-INV+0dB* is magnitude-flat to –0.03 dB at 28.93 Hz but its **phase** jumps from +32° to –116° in one step. The clamp constrains magnitude and leaves the phase discontinuity behind; step 8 then re-derives phase from a full-resolution magnitude that was never really flat. A hard clamp constrains amplitude, not bandwidth* — that part of the old diagnosis stands.)*

Step 8 — Minimum phase, second time

Do: Generate minimum phase on $F_l \rightarrow \text{LFilter}$. Likewise $F_r \rightarrow \text{RFilter}$.

dialog option	set to	why
Cal file effects	not included	a filter has no microphone. Including the mic calibration would bake the microphone's response into what you play as step 4: without it the transform corrupts the magnitude. This is the copy where it does the most damage as step 4
LF tail	yes — see the note below	
HF tail	no	

This is where the LF-tail error hurts most

Step 4's copies are damaged at 5–15 Hz, mostly *below* the correction band. Here the damage lands **inside** it — FL is unity below ~14 Hz, so the transform's sharpest edge is the band-limit transition itself, and that is where it rings:

$\backslash \text{Filter} \backslash$ — $\backslash \text{FL} \backslash$	5–15 Hz	15–20 Hz	20–25 Hz	25–40 Hz	40–225 Hz
max error	0.21	2.41	2.39	0.30	0.13 dB

In the exported WAV this resolves into a **–11.4 dB notch at 19.96 Hz**, identical on both channels to 0.01 dB — deterministic processing, not acoustics. It is 0.23 Hz wide at –6 dB, narrower than REW's 0.366 Hz display grid, so **the plot shows only a –4 dB dip and cannot resolve the core**. It is what fails the [R8](#) narrowest-feature and group-delay tests, both of which land at ~20 Hz.

Slope here is a real choice, not just conditioning. F1 is a filter, not a loudspeaker — it has no physical roll-off, and it is already unity below the band limit. A steep tail imposes a subsonic high-pass on the deliverable. That may be welcome, but decide it rather than inherit it: pick the **shallowest slope offered** if you want the filter left as designed.

The target is unaffected, because Step 5 builds it from LX/RX rather than the minimum-phase copies (that is the reason for the instruction). If you are rebuilding after fixing this, **LRrms+phavg**, the target shape and the calculated target level all stand — redo from Step 4 onward only.

Why a second time, when the divisor was already minimum phase. In exact arithmetic the quotient of a magnitude-only numerator and a minimum-phase divisor is minimum phase, so this ought to be a no-op. In practice it is not, for three reasons: the max-gain clamp and the one-octave band blend both perturb the phase; the division “uses windows that span the entire resulting impulse response as the result is typically not causal”; and the target has zero phase, which is not the same as having none. Measured on the deployed chain the step moved the 20 Hz phase from -41.6° to -2.3° .

What it guarantees: a causal, minimum-delay filter whose phase is exactly the Hilbert transform of its magnitude. That is the definition of the thing you meant to build.

Step 9 — Bake the crossover correction in, last

Do: Trace Arithmetic, **A times B:**

- FLX = A = **X801 (revised)**, B = LFilter
- FRX = A = **X801 (revised)**, B = RFilter

Why last, and why A = X801: keeping the room correction (LFilter) and the crossover correction (X801) as separate traces until the end means you can re-derive one without re-baking the other, and you can A/B the crossover correction by exporting with and without this multiply. And with X801 as trace A, the result inherits X801's window settings — i.e. none — which is correct for a finished filter.

Why the cascade is safe: LFilter is magnitude-derived minimum phase and contributes exactly nothing to crossover phase (§3); X801 has exactly flat magnitude and contributes exactly nothing to the magnitude correction. They are orthogonal. There is no double-correction to worry about.

Step 10 — Export

Do:

1. Trim IR to windows on FLX / FRX if you want to set the latency.
2. Export impulse response as WAV: **48 kHz, 32-bit float**. REW writes **131072 samples** — there is no tap count to set, the export length is fixed.
3. Name them **FLX-trimmed-48k.wav** / **FRX-trimmed-48k.wav**.

Trimming never changes the filter’s response. FLX and FLX-trimmed were compared bin by bin: identical to **0.0000 dB rms at every frequency**. All it does is move the impulse within the file — it sets latency, nothing else.

So **trim every filter you build**, including every rebuild: each pass through the chain produces a new FLX/FRX and each one gets trimmed and exported as normal. What trimming is not, is a **remedy**. If step 11 fails, re-trimming the same filter cannot change any of the three test results, because none of them depends on where the impulse sits in the file. The defect is upstream and only step 2 can reach it.

There is no “number of taps” in REW. rePhase has one; REW does not. The equivalent lever is the IR window of step 2, which is why that step matters so much. If you need a genuinely shorter filter, window the exported WAV outside REW with a **tapered** (Hann/Tukey) window — never a hard truncation, which produces ripple.

Step 11 — Accept or reject, before deploying

Do:

```
./drc_acceptance.py ../DRC-120.blue/FLX-trimmed-48k.wav \  
                    ../DRC-120.blue/FRX-trimmed-48k.wav --plot check.png
```

Exit status 0 = pass. Full description in [R8](#).

If it fails, do not deploy and do not post-process. Go back to step 2, lower the FDW cycles, and re-run the chain from step 3. A filter that fails these tests fails audibly.

Part IV — Reference

R1. The IR window dialog, field by field

Defaults as REW sets them on a full-range sweep:

field	default	what it does	change it?
Left window shape	Rectangular	taper applied before the reference time	no
Left width	500 ms	how much pre-arrival is kept	no — it does not affect resolution
Ref Time	~0	position of the window reference relative to $t = 0$	no — it is the IR peak, which is what the acoustic timing reference bought you
Right window shape	Rectangular	taper applied after the reference time	only on route B below
Right width	1000 ms	how much decay is kept	only on route B below
Add FDW	off, 15 cycles	the frequency-dependent Gaussian window	✓ ticked, 12 cycles

Frequency resolution = 1 / (right width). Settled by experiment.

REW’s help says the printed resolution corresponds to “*the current total window duration (left and right combined)*”. The dialog does not behave that way. Measured:

left	right	printed resolution	1/(L+R)	1/R
500 ms	1000 ms	1.00 Hz	0.67 Hz ×	1.00 Hz ✓
500 ms	500 ms	2.00 Hz	1.00 Hz ×	2.00 Hz ✓

The left width does not affect frequency resolution. It is not a resolution control; set the right width and leave the left alone.

Ref Time is not something you compute. REW puts $t = 0$ at the impulse peak, so Ref Time comes out at ~ 0 (0.042 ms on L.120.Blue — the residual offset of the peak from the sample grid). Move it only with the Impulse graph's *Set $t = 0$ at cursor* / *Offset $t = 0$* actions.

Route B — a fixed window instead of the FDW

Only if you want a flat Δf across the band. This gives the **wrong shape** — a constant Δf is far heavier regularisation at 4 kHz than at 50 Hz — so prefer the FDW unless you have a specific reason.

control	set to
Right shape	Tukey 0.5 — change this <i>before</i> the width
Right width	173 ms → prints 5.8 Hz $\approx 1/6$ octave at 50 Hz (300 ms → 3.3 Hz is a gentler first try)
Left shape / width	leave
Add FDW	off

total window	Δf shown	$\approx$ octave fraction at 50 Hz
1.5 s (default)	0.67 Hz	1/52 — effectively none
500 ms	2.00 Hz	1/17
300 ms	3.33 Hz	1/10
173 ms	5.78 Hz	1/6
86 ms	11.63 Hz	1/3

Do not do both. You would be regularising twice, and the printed resolution would no longer describe the data.

R2. Choosing a window shape

Only relevant on route B. Read it anyway — it explains why route A avoids the question.

(a) six shapes as REW applies them. (b) the same six in frequency — **this is the smoothing kernel** the response gets convolved with. (c) each applied to the real L.120.Blue impulse response at a 173 ms window.

[!] Textbook window figures do not apply to a half window

The familiar numbers (Hann -31 dB sidelobes, Blackman-Harris -92 dB) are for **symmetric** windows. REW's right window is truncated at its own peak, so the signal meets a step at the reference time however smooth the taper is, and that step dominates the kernel. Measured on the actual half-windows at 173 ms:

shape	-3 dB width	leakage outside main lobe	kernel
Rectangular	5.13 Hz	9.23 %	10 nulls — oscillates
Tukey 0.25	5.86 Hz	6.90 %	7 nulls — oscillates
Tukey 0.5	6.59 Hz	5.66 %	3 nulls — smooth
Hann	8.06 Hz	9.67 %	0 nulls — smooth
Hamming	6.59 Hz	11.72 %	8 nulls — oscillates
Blackman-Harris	9.52 Hz	12.75 %	0 nulls — smooth

Blackman-Harris is the worst of both worlds here — widest main lobe *and* highest leakage, the exact opposite of its symmetric-window reputation.

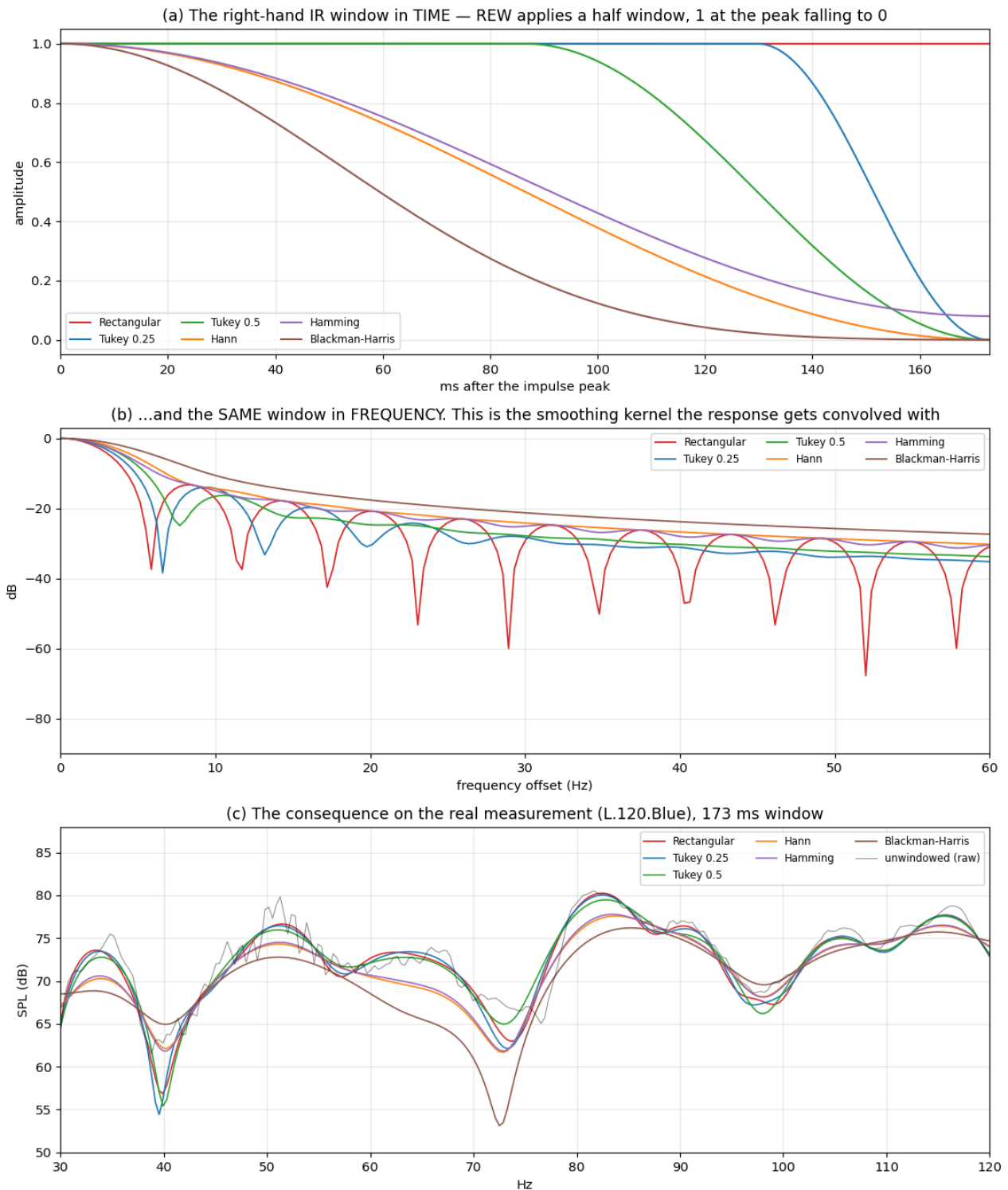


Figure 4: Window shapes in time, their frequency kernels, and the effect on the real measurement

Use **Tukey 0.5**, chosen on measurement rather than reputation: lowest leakage of the six and a smooth non-oscillating kernel, for only $1.28\times$ the resolution of rectangular. The distinction that matters is not sidelobe **level** — all six sit at -13 to -16 dB — but sidelobe **structure**. A rectangular kernel *rings*; a tapered kernel decays smoothly. Ringing in the kernel is ringing convolved into every feature of the response.

Why a time window is a smoothing kernel

Multiplication in time is convolution in frequency. Windowing the impulse response by $w(t)$ replaces $H(f)$ with $H(f) * W(f)$:

$$Y(f) = \int H(\nu) \cdot W(f - \nu) d\nu$$

Every output frequency becomes a **weighted average** of the input, with W supplying the weights — which is the definition of smoothing. A weighting function inside an integral operator is called its **kernel** (German *Kern*, “core”), and here **the kernel is literally $W(f) = \text{FFT}\{w(t)\}$** , which is what panel (b) plots. A window’s shape in time and its kernel in frequency are one object seen from two sides, which is why “choose a window length” and “choose a smoothing bandwidth” are one decision.

window in time	kernel in frequency	behaves like
rectangular, length T	sinc, main lobe $\sim 1/T$, -13 dB sidelobes	crude smoothing at $\Delta f = 1/T$, plus ringing
Tukey / Hann, length T Gaussian FDW, N cycles	wider main lobe, sidelobes $\ll$ Gaussian of width f/N	clean smoothing at $\Delta f \approx 1/T$ fractional-octave smoothing, $1/N$ per octave

One difference that matters. The smoothing menu convolves the **magnitude** — it averages $|H|$, so it can only make a null *shallower*. Windowing convolves the **complex** response, so contributions can cancel and a window can make a null *deeper*. REW’s help says the FDW has “an effect similar to applying a smoothing of the same octave fraction” — *similar*, not identical, and this is where it differs.

R3. Minimum phase — the two places you take it, and the one place you must not

#	applied to	producing	why
1	LX, RX	LX-MP, RX-MP	the divisor must be minimum phase. Divide by the raw measurement and the filter tries to invert the room’s excess phase: acausal, pre-ringing, valid at one microphone point
2	Fl, Fr	LFilter, RFilter	the filter must be causal. The clamp, the band blend and REW’s un-windowed division output all leave residual non-minimum-phase content
×	X801	—	never. Its magnitude is 0.00000 dB, so its minimum-phase copy is a unit impulse — you would delete the filter entirely

That last row is measurable, not rhetorical:

minimum-phase copy of X801.wav:

```

peak                1.000000  at sample 0
energy anywhere but sample 0  2.96e-14 %
```

Cal file effects: *included* at step 4 (you are modelling the acoustic response as measured, and the mic calibration is part of what that means), *not included* at step 8 (a filter has no microphone). Below 200 Hz a decent measurement mic is nearly flat, so this is a small effect — but the reasoning is asymmetric and worth following.

R4. How to invert: $\div$, $1/A$ and $1/|A|$

REW offers several operations that all look like “invert”. They are not interchangeable.

operation	result	outside its frequency limits	guards it offers	use it?
$A \div B$, $A = \text{Target}$, $B = \text{SUM-MP}$	Target $\div$ the mono sum	unity , blended over one octave	maximum gain, regularisation	✓ the common filter , 20–80 Hz (§11)
$A \div B$, $A = \text{Target}$, $B = \text{LX-MP}$	Target $\div$ one channel	unity , blended over one octave	maximum gain	✓ the per-channel filter , 80–225 Hz
$A \div B$, $A = \text{Target}$, $B = \text{LX-MP}$	Target $\div$ measurement	unity , blended over one octave	maximum gain, regularisation	the older single-division form; deepens the 45–56 Hz mono cancellation
$1/A$ on LX-MP	flat at a chosen level	unity , blended over one octave	maximum gain, target level, exclude notches	only if you want a flat target and no house curve
$1/ A $ on LX-MP	magnitude inverse	—	—	× produces a linear-phase result — symmetric pre-ring, and no phase correction at all
$A \div B$ with $B = \text{LX}$	inverts magnitude and room excess phase			× acausal, pre-rings, one-point-only

Why $A \div B$ and not $1/A$: the division takes a **target curve**, so the house curve of step 5 comes along for free. $1/A$ takes only a scalar *target level*, so it can only aim at flat. If you are happy with flat-at-the-average across 20–225 Hz, $1/\text{LX-MP}$ in one step is perfectly respectable and gives you the **exclude notches** option, which $A \div B$ does not have.

The one thing $1/A$ has that division does not is that notch-exclusion checkbox, which drops notch-like features out of the inversion entirely rather than merely limiting their boost. If a null survives step 2, that is the one remaining lever — at the cost of the target curve.

Never $1/|A|$. REW’s help: “*The $1/|A|$ and $1/|B|$ operations produce a linear phase result.*” A linear-phase filter is symmetric: half its energy arrives **before** the impulse. That is the opposite of what step 8 is for.

R5. The three boost guards

guard	belongs to	mechanism	effect on bandwidth	set it to
Maximum gain	division, inversion	hard clamp at a level	none — a clamp is a discontinuity, and a discontinuity is a narrow feature	0 dB — gives a cut-only filter
Regularisation	division	a floor under the divisor, as a % of its average level	none	0 , unless max gain is above 0 dB
Exclude notches	inversion only	drops notch-like features from the inverse	removes them entirely	off

All three limit boost. None of them limits bandwidth. So none of them can prevent the failure in §5, which was a *dip* — 0.73 Hz wide, 3.5 dB from rim to bottom — in a filter that never boosted at all. **The FDW is not a backstop to these controls; these controls are a footnote to the FDW.**

With 12 cycles applied at step 2 the 38.3 dB null is already an 11.7 dB dip and the deepest point in the whole 20–225 Hz divisor is 13.7 dB below its average level, so the clamp barely engages and there is nothing left for a floor to sit under.

REW’s own framing of regularisation:

“Applying regularisation limits the boost which occurs when the divisor becomes very small (or zero!), such as where the divisor has notches in its response, and so produces a more stable and manageable result. The maximum gain corresponding to the regularisation percentage is shown next to the control, at 25% (the maximum setting) no gain is allowed.”

R6. Getting the room’s decay figure

You need one number to choose the FDW cycles: how long this room actually decays in the 40–100 Hz region. **No single method gives it below ~150 Hz.**

tool	gives	trustworthy where	use it for
RT60 (T20 / T30 / EDT / T _{opt})	numbers per band	above Schroeder only (~166 Hz)	mid/high-band RT, treatment work
Waterfall	3D ridges	qualitative	<i>which</i> modes ring; before/after
Spectrogram	2D map, colour = level	qualitative, $\pm 2\times$	<i>how long</i> — the referee
Peak bandwidth	numbers per mode	below Schroeder, if cross-checked	the figure you actually use

Route 1 — from peak bandwidth. Since $T60 = 2.2 \cdot Q / f_0$ and $Q = f_0 / \Delta f$, the frequency cancels:

$$T60 = 2.2 / \Delta f, \text{ where } \Delta f \text{ is the } -3 \text{ dB bandwidth of the peak, in Hz.}$$

Decay depends *only* on bandwidth. A 5 Hz-wide peak decays in 440 ms whatever its frequency. Measure Δf off the **unsmoothed** trace with the cursor.

Its blind spot: a single-point magnitude measurement **cannot distinguish a genuine high-Q mode from a narrow interference peak**. Both are narrow, and the formula converts either into a decay time. On this dataset it returns 2215 ms at 53.8 Hz — physically impossible in 58 m³ — which is the §5 trap seen from the frequency side.

Route 2 — RT60 graph. Standard, but meaningful only **above** Schroeder. Below it the field is modal rather than diffuse, “reverberation time” is not well defined, and the analysis band filter has its own floor: a 1/3-octave filter at 25 Hz rings 205 ms unaided. Numbers from this route below ~150 Hz are not trustworthy.

Route 3 — spectrogram, as the referee. It cannot give a number, but it answers the one question the others cannot: *how long does energy at this frequency actually persist?* Here, the 40–100 Hz band fades in roughly **300–400 ms**.

Why no view can give you a number — the same law, again. Both waterfall and spectrogram are built on an analysis window of length W , and that window sets *both* axes: frequency resolution $\approx 1/W$, time smearing $\approx W$. To resolve a 5 Hz-wide mode at 50 Hz you need $W \geq 200$ ms; but a 200 ms window cannot resolve a decay much shorter than 200 ms, and the decay you are measuring is ~400 ms. There is no setting that gives both. $\Delta f \cdot \Delta t \approx 1$ (§4), now limiting the instrument you would use to check the instrument.

These views are reliable to a **factor of about two** — enough to reject 2200 ms, nowhere near enough to separate 300 from 400 ms. Referee, not instrument.

Reading a spectrogram without fooling yourself:

- Analysis window $\approx$ **300 ms** for bass work: long enough to separate modes ~3 Hz apart, short enough not to invent decay.
- Level range so the floor sits **35–40 dB** below the peak. Too wide and everything looks like it rings forever; too narrow and real decay vanishes.
- Read decay as *where a streak fades relative to where it started*, with the cursor — not by eye off a screenshot.
- **Ignore everything below ~20–25 Hz.** The sweep has little energy there and the window is many cycles long; the smear is the analysis, not the room.

So: route 1 for the numbers, route 3 to reject the impossible ones. Any peak whose bandwidth implies a decay far longer than the spectrogram shows is interference, not a mode, and **must not be corrected**. Cross-checked that way, the credible decays here are **300–400 ms** in the 40–100 Hz band, which is the table in §8.

Applying the same arithmetic to the features the September filter built:

feature in FLX	Q	source resonance it implies	verdict
28.93 Hz	40.4	T60 = 3.07 s	impossible
51.27 Hz	62.7	T60 = 2.69 s	impossible
81.67 Hz	11.9	321 ms	plausible
116.46 Hz	15.7	297 ms	plausible

Independent proof that the two worst features were never room modes — and the answer to “does the window throw away something real”: for exactly those features, no.

R7. X801 — what it is, and do not redesign it

X801.wav linearises the two Nautilus 801 crossovers and nothing else. Measured directly from the file:

	value
magnitude, 5 Hz – 23 kHz	–0.00000 to +0.00000 dB — a perfect all-pass
group delay fit, low crossover	375.4 Hz, Q 0.706
group delay fit, high crossover	3945.5 Hz, Q 0.686
residual of the two-all-pass fit, 25 Hz – 15 kHz	0.0076 ms rms
fitting the high crossover alone	26× worse
relative group delay, 20 Hz vs treble	–1.30 ms
length	131072 samples, 32-bit float, 48 kHz, centred

It contributes **nothing** to magnitude and therefore nothing to low-frequency ringing. It is ready to use as-is.

[!] X801.rephase on disk does not match X801.wav

The saved rePhase project has an **empty filter type** on the 355 Hz row, but the WAV measurably corrects both crossovers. **Open that project and re-export and you will silently lose the 350 Hz correction — which carries ~95 % of the group delay.**

The WAV is the artifact of record. If it must ever be rebuilt: low row at ~355–375 Hz, LR 24 dB/oct ($Q \approx 0.707$), which is what the fit says is actually in there.

How it is designed in rePhase: the *filter linearization* tab, entering the two crossover frequencies and their topology. That is the whole design, and it is the correct minimal one.

Not to be confused with Xo801.wav

DRC-185/Xo801.wav is the *same two crossovers* — they agree above 350 Hz to **0.0051 ms rms** — **plus** an extra low-frequency term: a 4th-order high-pass at **20.66 Hz**, i.e. a bass-reflex box alignment. It is not wrong, it is a different **scope**.

Use the crossover-only X801. Xo801's cost is genuinely small (1.6 ms of pre-ring at –40 dB), but its 20.7 Hz figure is a model fitted to the filter itself and never checked against a nearfield measurement of a real 801 — and below 100 Hz the room contributes several times more phase than the ~22 ms it corrects. Low cost, unverified benefit. Restore it only after measuring the speaker's rolloff nearfield.

Minor open point: the N801 crossover is nominally third-order, whose sum is an all-pass of $Q = 1.0$, while the fit says $Q \approx 0.706$ (LR4-like). If the real acoustic sum is $Q = 1.0$ at 350 Hz, about 0.6 ms of rotation remains uncorrected — at or under the Blauert & Laws audibility threshold. A refinement, not a defect.

R8. Acceptance tests

drc_acceptance.py, in this directory. Run it on the WAV BruteFIR actually loads. Exit status 0 = pass.

```
./drc_acceptance.py ../DRC-120.blue/FLX-trimmed-48k.wav \
../DRC-120.blue/FRX-trimmed-48k.wav --plot check.png
```

test	pass	September FLX
sharpest feature below 200 Hz	Q of 12 or less	Q 73 ×
group delay excursion, 20–200 Hz	< 10 ms	80 ms ×
gated-tone tail, any tone below 200 Hz	< 100 ms	1348 ms ×
magnitude, DC ... 20 Hz	flat, no correction	ok

[!] **The first test used to be in FFT bins. That was wrong**

Bin spacing is f_s / n , so a bin-based threshold **passes or fails the same filter depending on how long the exported file is**. Trimming one of these filters from 262144 to 131072 taps changed its verdict without changing the filter: the band-edge feature measures **9.07 Hz** wide in one and **9.37 Hz** in the other — the same feature — but 49.6 bins against 25.6, because each bin doubled in width.

The criterion is now **$Q = \text{centre frequency} \div \text{bandwidth}$** , which is dimensionless and length-independent. The threshold follows from §8: an N-cycle FDW caps fractional bandwidth at $1/N$, so a divisor windowed at 8–12 cycles cannot legitimately yield a feature sharper than about 12.

Calibrated against this project’s history:

	width	at	Q	
Sept 2025 deployed filter	0.30 Hz	28.9 Hz	73	× rang 1348 ms
no-LF-tail notch	0.27 Hz	20.0 Hz	73	×
74 Hz SBIR shoulder, 12-cycle build	5.78 Hz	73.2 Hz	12.7	× marginal
8-cycle build	13.1 Hz	99.6 Hz	7.6	✓
20 Hz band-edge transition	9.37 Hz	25.6 Hz	2.7	✓

Note this also changes *which* feature is reported: “sharpest” now means largest $f_0/\text{bandwidth}$, not smallest bandwidth. A 13 Hz feature at 100 Hz is sharper in the sense that matters than a 9 Hz one at 25 Hz.

Bins are still printed, for reference only.

The script uses a **matched-latency control** for the gated-tone test and applies a **4-period floor** — a 40 dB decay of a 28.7 Hz tone cannot resolve faster than ~139 ms, so demanding less would measure the envelope estimator rather than the filter. Verified behaviour: **X801.wav passes** all three; **FLX-trimmed-48k.wav fails** all three.

What to run it on: **FLX-trimmed-48k.wav / FRX-trimmed-48k.wav**. Also run it on LFilter/RFilter if you want to separate room-correction ringing from anything the crossover filter contributes — **X801.wav passes** on its own, so any failure is the room part.

By hand, if you prefer, import the filter as a measurement (File → Import → Impulse Response) and look at **Group Delay** first. It is one graph and it caught every failure in this dataset:

	28.93 Hz	51.27 Hz	79 Hz	116.5 Hz	max 20–200 Hz
FLX (L)	– 105 ms	– 97 ms	–0.9	–35	+80 ms @ 29.3 Hz
FRX (R)	+0.7	+0.8	– 65 ms	–32	+52 ms @ 75.4 Hz

A correction filter should show group delay of a **few milliseconds** below 200 Hz. Tens of milliseconds means you have built a resonator.

Seeing all three tests in REW, and comparing two filters

Every number the script produces has a REW view behind it. Import the WAV with File → Import → Impulse Response and it becomes a measurement like any other.

REW view	what it shows	script test
Waterfall (CSD) or Spectrogram	how long each frequency keeps ringing	gated-tone tails
Group Delay	resonators, as GD peaks	group delay excursion
SPL, Smoothing: None	narrow features	narrowest feature
Step Response	pre-ringing and overshoot	pre-peak energy

Waterfall settings for a filter (not for a room — the defaults assume a room): time range ~**400 ms**, rise time **100 ms**, **40 dB** range. Anything still standing at the back of the plot is what the script calls a tail.

To compare two filters, import both and use the **Overlays** window. All SPL and Group Delay overlay directly, which is the fastest way to see what a rebuild changed. Waterfalls do not overlay — flip between them with the measurement selector.

Thresholds, so you can judge audibility rather than pass/fail

The script’s thresholds are deliberately conservative — they are there to catch construction faults, not to predict audibility. These are the numbers for the second question. Both are approximate and the literature disagrees by roughly a factor of two.

Group delay (Blauert & Laws; Møller):

20 Hz	50	100	200	500	1 kHz
~45 ms	~32	~24	~18	~12	~6 ms

Modal ringing: audibility for bass decay is usually placed around **200–300 ms** for normal program material.

Applied to this project’s filters:

	worst group delay	verdict	worst tail	verdict
April build	–104.6 ms @ 28.9 Hz	~2.5× over threshold	>2000 ms @ 28.7 Hz	far over
8-cycle build	+17.2 ms @ 31.9 Hz	under (~38 ms there)	114 ms @ 79 Hz	under

The 8-cycle build still **fails** the script on group delay — 17.2 ms against a 10 ms criterion — while sitting **below the audibility threshold** at the frequency where it occurs. That is the criterion doing its job: it is a build-quality gate, not a verdict on the sound. Read a failure as “look at this”, not as “this is audible”.

What the guards cost

Rebuilding the left filter from its own magnitude with LF regularisation:

gated note	as deployed	1/6 oct < 200 Hz	1/3 oct < 200 Hz
28.7 Hz	1348 ms	1	1
51.2 Hz	787	72	16
79 Hz	1209	72	34
81.6 Hz	511	46	35
116.5 Hz	264	45	30
145.5 Hz	465	30	9
narrowest feature < 200 Hz	0.8 bins	31.5 bins	126.7 bins
change to the correction curve	—	1.29 dB rms	1.52 dB rms

1.29 dB rms of correction accuracy buys the removal of 1.3 seconds of ringing. That is the trade this whole procedure makes, quantified.

How the gated-tone test works

Worth understanding, because it is the one test that maps directly onto something you can hear. For each of ten tones — 28.7, 40, 51.2, 63, 79, 100, 116.5, 128.2, 145.5, 180 Hz, the room’s modal frequencies — the script:

1. builds a **1-second sine**, with 5 ms raised-cosine ramps at both ends so that switching it off does not itself click;
2. appends **2 seconds of silence** — the note stops dead;
3. **convolves it with the filter**;
4. takes the **Hilbert envelope**, reads the steady-state level from the 300 ms before note-off, and times how long the envelope takes to fall **40 dB** below it.

That time is the **tail**. It tests **the filter alone**, not the filter in the room: whatever it rings is *added* to the room’s own decay.

The **control** column is a pure delay pushed through the identical chain. It should ring for zero time, so whatever it reports is the measurement’s own floor — the envelope estimator needs several cycles to resolve a 40 dB drop, which takes longer as frequency falls. Hence controls of ~124 ms at 28.7 Hz falling to 3 ms at 180 Hz. The limit is $\max(100 \text{ ms}, 3 \times \text{control}, 4000/f_0 \text{ ms})$, the last term being four periods.

[!] A single tone length is not a measurement

The gate closes at whatever phase the tone happens to be in, and the envelope of a low tone ripples at $2f_0$, so the 40 dB crossing lands on either side of a ripple lobe. On an **unchanged** filter, 28.7 Hz returned 80, 82, 149, 217, 259 or 85 ms depending on a 2 % change in tone length.

The script therefore reports the **median over nine lengths** and an **IQR spread column**, and flags any entry whose spread exceeds half its median as **noisy**. **Entries flagged noisy are not evidence in either direction** — read the spread before believing any tail.

This is not academic. The single-length version made the 8-cycle rebuild above look *worse* than the 12-cycle one, on two filters that are magnitude-identical at 28.7 Hz to 0.04 dB and therefore cannot differ there. The **control** was hit harder than the filter: it drew 0 ms at 28.7 Hz while reading 94 ms at 40 Hz. A pure delay’s floor must grow monotonically as frequency falls, so that zig-zag was the tell — and because the limit is $3 \times \text{control}$, a control that collapses to zero drops the 28.7 Hz limit from 373 ms to 139 and fails filters that pass.

If the controls in your output are not monotonic in frequency, do not trust the tails.

Worked example: R8 blinded by processing, then not

The 2026-08-12 rebuild is the clearest demonstration of what these tests are for. The same filter, built twice from identical measurements, differing only in the **LF tail** of step 4 and step 8:

test	No LF tail	LF tail at 16 Hz
narrowest feature < 200 Hz	× 1.49 bins @ 19.96 Hz	✓ 31.56 bins @ 73.24 Hz
group delay, 20–200 Hz	× +257.4 ms @ 20.32 Hz	× +24.1 ms @ 74.52 Hz
gated tones failing	5 , worst >2000 ms @ 28.7 Hz	2 , worst 228 ms @ 63 Hz

Two things to take from it.

First, the frequency response barely moved — the delivered mono sum changed by +0.01 to +0.12 dB in every band. All three tests improved by an order of magnitude while the thing you would look at in REW stayed put. That is the entire argument for running time-domain acceptance tests at all: **the defect was invisible on the magnitude plot**, and REW’s own 0.366 Hz grid could not resolve it even in principle (§ step 8).

Second, look at where the failures moved to. Every remaining one sits at **63–92 Hz**, and the filter there reads:

68 Hz	71	73	74	76	79	85	92
–2.68	–0.88	–0.04	–0.12	–1.51	–6.22	–10.37	–8.09 dB

That unity plateau at 73–74 Hz between two cuts is the front-wall SBIR null being clamped to unity by max gain 0 — the mechanism [step 6](#) describes, with geometry predicting the null at 75.3 Hz. Ten decibels of swing in half an octave is a high-Q feature, and a high-Q feature is a long tail: 63 and 79 Hz are the **shoulders of that null**.

So R8 stopped reporting an artifact and started reporting the one real acoustic problem in the room. **A failing test that has moved onto a feature you can point at in the geometry is a different kind of failure from one sitting on a band edge.** The first is a decision about how hard to correct; the second is a bug.

R9. Does the source correlation invalidate §11?

§11 divides by the **vector sum**, which is the response you get when both speakers reproduce *the same waveform*. Real music below 80 Hz is not necessarily the same in both channels, so the question is fair: is the divisor describing a situation your recordings are not in?

Correlation, in one paragraph

Two speakers can only cancel each other if they are **playing the same thing**. The 45–56 Hz cancellation is destructive interference between the left and right arrivals — which requires the two arrivals to carry the same waveform. Feed the speakers unrelated signals and there is nothing to cancel; the two just add in energy, like two people talking at once. The measure of “same waveform” is the correlation **r** between the channels, band-limited to the region of interest:

r	the two channels contain	consequence at the seat
+1	the identical waveform	the speakers cancel fully — the measured null is real
0	unrelated waveforms	no cancellation at all — powers simply add
−1	the same waveform, inverted	cancellation on the mono sum is at its worst

Note r runs from **−1 to +1**, and negative values do occur in real recordings. Note also that correlation is **not** the same as balance: equal energy in both channels with $r \approx 1$ gives a solid centre image, while equal energy with $r \approx 0$ gives diffuse bass with no location at all.

What the room measurement assumes

A sine sweep is perfectly correlated by construction. So the vector sum shows the $r = 1$ case — **the deepest the null can ever be**. The physically correct divisor for source correlation r is

$$|H_L|^2 + |H_R|^2 + 2r \cdot \text{Re}(H_L \cdot H_R^*)$$

which is the vector sum at $r = 1$ and the power sum — essentially the RMS average — at $r = 0$.

What is actually on the discs

Measured over 71 albums of this collection, one track each, band-limited to 20–80 Hz:

	median r	energy-weighted mean	range
all tracks	+0.65	+0.46	−0.57 ... +0.99
tracks with >3 % of energy below 80 Hz	+0.50	+0.44	−0.57 ... +0.99

Fully spread — Mendelssohn organ discs at 0.93–0.99, Shostakovich’s 1st and 15th at **−0.57**. At $r \approx 0.45$ the correct divisor sits between the two extremes, and it differs from the vector sum by **2.9 dB rms across 45–56 Hz, worst 4.5 dB**, while agreeing to within 0.25 dB everywhere else.

And yet it changes nothing — because of the clamp

That 2.9 dB is an error in the **divisor**, not in the **filter**. Carried through to what the filter does:

Hz	target	divisor r=1	divisor r=0.45	cut applied, r=1	cut applied, r=0.45
45	71.5	67.2	67.9	0.00	0.00
50	71.1	67.7	70.4	0.00	0.00
56	70.6	63.9	68.6	0.00	0.00
71	69.3	71.3	71.7	−2.06	−2.45
80	68.6	74.9	75.3	−6.33	−6.75

Across 45–56 Hz the divisor lies **below** the target on either assumption — by 3.5 dB at 50 Hz with $r = 1$, still 0.7 dB with $r = 0.45$. Both ask for a **boost**, and **max gain 0 dB clamps every boost**. The clamp is engaged on 100 % of bins in that band with both divisors, so the filter is identical.

The only real difference is **≈0.4 dB more cut at 71–80 Hz** — below session repeatability, and not worth a rebuild.

The general lesson, which is the reason this appendix exists

An error in the divisor is only an error in the filter **where the filter is acting**. Below 80 Hz this design spends most of its time clamped, so discrepancies there are absorbed rather than propagated. The same reasoning retires the narrow dips in [step 6](#) and the deep null at 74 Hz. **Before acting on a divisor discrepancy, carry it through $\text{min}(\text{Target} - \text{divisor}, 0)$ and see whether it survives.** It usually does not.

(Caveat worth keeping: one 90-second excerpt per album is decent breadth and thin depth, and r varies within a work as much as between works. The conclusion above is robust because the clamp absorbs the whole range of r , not because 0.45 is a precise figure.)

R10. Troubleshooting

symptom	cause	fix
Woofers keep moving after the music stops	narrow features in the correction curve → high-Q resonators	FDW at step 2. Verify at step 6
Group delay spike of tens of ms below 100 Hz	same	same
Step 6 fails — features still 1–2 bins wide	the FDW never reached the data	check it was applied to L/R, that Apply Windows was pressed, and that step 3 was done after step 2
Filter is not flat above 300 Hz	the division's frequency limits were not set	step 7 — set 20 Hz and 225 Hz
Filter has energy <i>before</i> its peak	$1/ A $ was used, which is linear phase; or the divisor was LX not LX-MP	steps 4 and 7
Crossover phase unchanged after correction	expected — a minimum-phase inversion cannot see an all-pass	step 9; see §3
Applying 1/6 octave smoothing changed nothing	trace arithmetic ignores the smoothing menu	§9 — use the FDW
Correction sounds right at the mic, wrong one seat over	single-point correction of position-specific features	more positions (§6), or fewer FDW cycles
Changed the window, the filter did not change	derived traces are snapshots, not live views	redo the chain from step 3
Deep null got <i>deeper</i> after the FDW	the FDW convolves the complex response; that null is early-arrival, not late	it is real — do not try to fill it

R11. Glossary

FFT bin. A measured response is a list of values at evenly spaced frequencies. The spacing here is $48000 \div 131072 = \mathbf{0.3662\ Hz}$. Nothing narrower than one bin can exist in the data, so a “2-bin feature” is 0.73 Hz wide — the finest structure the format can represent, and a guarantee that no averaging of any kind was applied.

Kernel. The weighting function inside an integral operator — the shape of the weights used when averaging neighbouring bins. For a time window, the kernel is literally the FFT of the window. Its width is what “1/6 octave” specifies. From German *Kern*, “core”. A kernel narrower than one bin changes nothing.

Minimum phase. A system whose phase is the Hilbert transform of the log of its magnitude — the unique causal system with a given magnitude and the least possible delay. Invertible with a causal, stable filter, which is why the whole procedure passes through it.

Excess phase. What is left after factoring out the minimum-phase part: magnitude exactly 1, pure delay and rotation. Contains the propagation delay, the room's reflections, and the crossover all-pass.

Q and decay. A resonance of quality factor Q at f_0 decays 60 dB in $T_{60} = 2.2 \cdot Q / f_0$, and 40 dB in $t_{40} \approx 1.47 \cdot Q / f_0$. Inverted: to keep ringing under T milliseconds you need $Q < T \cdot f_0 / 1466$.

target	at 50 Hz	at 80 Hz	at 120 Hz
ring < 100 ms	$Q < 3.4$	$Q < 5.5$	$Q < 8.2$
ring < 50 ms	$Q < 1.7$	$Q < 2.7$	$Q < 4.1$

Note that Q is not a setting anywhere in this procedure. In the parametric / Auto EQ path it is an explicit control. In the inversion path there is no filter list and no Q field — the filter is a spectrum, and Q is an *output*, determined entirely by how narrow the features in your correction curve are. You control it only through the FDW. **The Q rule is an acceptance test, not a setting.**

Gated-tone tail. Play a steady sine, stop it abruptly, measure how long the output takes to fall 40 dB. The direct numerical version of “the music stops and the woofer keeps moving”. Unlike band-filtered decay it needs no analysis bandpass, so it has no artifact floor to subtract.

Schroeder frequency. $2000 \cdot \sqrt{(T60/V)}$ — above it the room is diffuse and statistical, below it modal and position-specific. Here, ≈ 166 Hz. It is the frequency above which a single-point measurement starts to mean something.